

Small Shocks and Exit Cascades: the Role of Non-Exclusive Competition and Outside Options*

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Abstract

When do small deteriorations in market quality trigger sudden, widespread collapses in trade? Existing results show that markets are robust to small shocks under exclusive contracting, while collapse is possible under knife-edge conditions in non-exclusive contracting with linear preferences. We ask when collapse is a general phenomenon. In the non-exclusive environment of [Attar et al. \(2021\)](#), we first show markets are generically robust: participation is governed by marginal conditions unaffected by another type's exit, and active layers survive sufficiently small shocks. This robustness breaks down once outside options are generalized beyond autarky. We establish conditions for exit cascades: non-exclusive contracting, (partial) pooling, and outside options above the no-trade utility. A single exit then destroys the cheap shared layer that its presence provided to every remaining type, triggering a cascade that shuts down the entire market. The separating equilibria of [Attar et al. \(2014\)](#) are immune to such cascades even with generalized outside options due to the separation of contract pricing across types.

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1 Introduction

Financial crises are often sudden: credit markets that function smoothly for years can seize within weeks as risk accumulates and adverse selection worsens. As [Krishnamurthy and Muir \(2025\)](#) and [Benmelech and Bergman \(2018\)](#) document, small deteriorations in underlying fundamentals have historically preceded large, discontinuous collapses in market activity. Understanding the mechanisms that convert gradual risk accumulation into abrupt market shutdowns is therefore a central question in financial economics.

This paper asks: when can a small deterioration in the quality of market participants cause trade to collapse entirely? The canonical framework of [Akerlof \(1970\)](#) provides one answer under exclusive contracting—when there is a single market-clearing price, adverse selection can unravel trade entirely. But in many important markets—insurance, over-the-counter derivatives, collateralized debt and loan obligations—agents are free to trade simultaneously with multiple counterparties. This *non-exclusive contracting* environment, which better describes these markets, has a richer equilibrium structure, and it is far from obvious whether and when the Akerlof mechanism survives. Across the canonical contracting environments we study, our results isolate the combination of non-exclusive contracting, shared-price partial pooling, and outside options above autarky as the ingredients that generate non-knife-edge exit cascades. In the layered environment of [Attar et al. \(2021\)](#), these forces can turn the exit of a single lowest-cost type into a cascade and complete market shutdown.

Two prior results are essential building blocks for understanding our contribution. [Azevedo and Gottlieb \(2017\)](#) develop a general model of perfectly competitive adverse selection under exclusive contracting that accommodates very general preferences, contracting spaces, and outside options. They show that their equilibrium is robust to small changes in fundamentals: outside of knife-edged cases, the “only way to produce large changes in equilibrium predictions is to considerably move average cost or demand curves.” This establishes that exclusive contracting, by itself, cannot generate sudden market shutdowns—the mechanism must lie elsewhere. [Attar](#)

et al. (2011) provide the first hint of where. In a non-exclusive version of Akerlof (1970) with linear preferences and capacity constraints, all trades collapse into a single pooling price, and this structure can generate knife-edged market unraveling: a small deterioration in fundamentals can lead an equilibrium with positive trade to one with none at all. However, the result is fragile and special—it requires linearity, a single pool, and the precise conditions under which the demand curve is tangent to the average cost curve.

Our main results complete this picture. First, we show that neither of these special features—exclusive contracting on one side, linear single-pool on the other—is what drives market fragility. In the general non-exclusive contracting environment of Attar et al. (2021), which features non-linear pricing and multiple types trading across many price layers, small cost increases *do not* generate exit cascades or market shutdowns. Even as the best (lowest-cost) type exits and market prices deteriorate, participation is governed by marginal conditions that another type’s exit leaves unchanged: remaining agents lose inframarginal utility but retain their incentives to trade. This extends the robustness intuition of Azevedo and Gottlieb (2017) to the non-exclusive contracting setting, and shows that non-exclusivity alone is insufficient. Second, once we generalize outside options—allowing agents to exit to utility levels strictly above autarky—small cost increases *can* cause a cascade of exits and complete market shutdown in Attar et al. (2021). The exit of the best type removes the cheapest shared layer from every remaining agent’s opportunity set, lowering total utility and potentially pushing agents past their outside option threshold. One exit begets the next. Third, this cascade mechanism requires partial pooling in equilibrium. In the simultaneous contracting environment of Attar et al. (2014)—which admits only separating equilibria for non-zero trades—outside options cannot generate cascades even when generalized, because the exit of one type relaxes incentive compatibility for the remaining types, weakly improving their terms of trade. Fourth, economies with more *coarse* partitions of types are less prone to cascades: when adjacent types are pooled into a single category, the relative mass of each partition is larger, requiring a bigger shock to trigger the initial exit that cascades.

The key to understanding why the equilibrium structure of [Attar et al. \(2021\)](#) generates cascades—while [Attar et al. \(2011\)](#) only does so in a knife-edged way, and [Attar et al. \(2014\)](#) not at all—lies in how types are linked through shared prices. In [Attar et al. \(2021\)](#), trades are organized as recursive layers along a convex-market tariff: all types share the first price layer, types 2 through n share the second, and so on. When one type exits, the cheap layer it supported disappears from the choice set of every remaining type. This *shared-layer exposure* creates strategic complementarity in exit decisions that is absent from both the single-pool linear model of [Attar et al. \(2011\)](#) (where pooling is too coarse to differentiate types) and from separating environments. Outside options then provide the threshold that converts deteriorating terms of trade into an actual exit, resulting in additional discrete decline in terms of trade. In this sense, our results identify (partial) pooling with non-exclusivity and generalized outside options as the precise combination necessary for fragility—generalizing and sharpening the [Attar et al. \(2011\)](#) knife-edge insight.

The model requires no institutional details, dynamic mechanisms, or sentiment-driven runs: just a single-crossing property on preferences and a monotonicity condition on costs. This parsimony makes the framework widely applicable. CDO and CLO markets, interbank lending, insurance, and OTC derivative markets all feature non-exclusive contracting and partial pooling. For example, CDO/CLO issuance collapsed discontinuously during crises even as underlying default rates increased only modestly.

Finally, our results reveal a surprising dual role for outside options. In [Attar et al. \(2014\)](#), where equilibrium is separating, outside options can actually *open* markets: if one type’s outside option is sufficiently attractive that it exits voluntarily, the adverse selection problem disappears and the remaining type can trade at its type-specific price—an endogenous screening device in the spirit of [Board and Pycia \(2014\)](#). It is only when partial pooling links agents’ terms of trade that outside options flip from market-opening to market-closing.

The paper proceeds as follows. Section 2 examines two special cases that motivate our general analysis. Section 3 presents the general framework of [Attar et al. \(2021\)](#).

Section 4 establishes that absent outside options, active markets are robust to small cost increases. Section 5 introduces generalized outside options and derives necessary and sufficient conditions for a cascade of exits—partial pooling combined with outside options above autarky. Section 6 shows that the separating equilibria of Attar et al. (2014) are immune to cascades even with generalized outside options, isolating partial pooling as the essential ingredient for market fragility. Section 7 includes the empirical implications our model generates. All proofs are relegated to the appendix.

Relation to the Literature. Our paper sits at the intersection of non-exclusive contracting under adverse selection and market unraveling conditions.

We build on the non-exclusive contracting literature (Pauly, 1974; Jaynes, 1978; Hellwig, 1988; Glosten, 1994; Attar et al., 2011, 2014, 2021; Dubey and Geanakoplos, 2019), where agents trade in layered contracts. Our contribution extends Attar et al. (2011)’s knife-edge unraveling result from linear preferences and single pooling to general preferences and multi-layered equilibria. While Attar et al. (2014, 2021) consider general preferences, they do not establish unraveling conditions. We derive the general conditions under which small changes in fundamentals cause market shutdown.

Our paper is the non-exclusive counterpart to Azevedo and Gottlieb (2017), who show that exclusive contracting markets are resilient to small shocks. We confirm this robustness extends to non-exclusive settings absent outside options, but identify when it fails: the combination of pooling equilibria and generalized outside options. This sharp characterization exploits cross-subsidization absent in exclusive contracting, which creates strategic complementarity in exit decisions. Unlike dynamic or belief-driven mechanisms (Kurlat, 2013; Malherbe, 2014; Asriyan et al., 2019), ours operates in a static, belief-free framework.

We show Hendren (2014)’s result—that markets with a probability-1 type must unravel under exclusive contracting—is sensitive to contracting form: under non-exclusivity, such types need not cause unraveling. Unlike Levy and Veiga (2021), we show both Rothschild-Stiglitz and Akerlof unraveling occur even with bounded costs, with exit decisions being strategic complements or substitutes depending on outside

options.

Recent applications ([Asriyan and Vanasco, 2024](#); [Kurlat, 2016](#); [Bigio and Shi, 2025](#)) feature semi-pooling equilibria—precisely the environments our results identify as vulnerable to exit cascades.

2 Special Cases with Market Shutdowns

Before developing our formal analysis, it is useful to understand how market unraveling arises—and fails to arise—in existing environments. We begin with [Azevedo and Gottlieb \(2017\)](#), who develop a general model of perfectly competitive adverse selection under exclusive contracting. Their framework is highly flexible: preferences and the contracting space are general, equilibrium can be separating, pooling, or partial pooling, and outside options are permitted. Though robustness is not their main focus, they establish in Section 4.3 that equilibrium is resistant to small changes in fundamentals—“the only way to produce large changes in equilibrium predictions is to considerably move average cost or demand curves.” Thus, exit cascades and complete market unraveling are not generic features of exclusive contracting environments.¹

This motivates the study of non-exclusive contracting environments, where agents trade simultaneously with multiple counterparties. We first consider [Attar et al. \(2011\)](#), who establish a general non-exclusive contracting framework with linear preferences and capacity-constrained agents. These assumptions generate a strategic foundation for an [Akerlof \(1970\)](#)-like equilibrium: all trades collapse into a single pool at a price equal to the average quality of the good, and agents either trade their entire endowment or nothing at all. Within this single-pool structure, market unraveling is possible at a knife-edge: when equilibrium gains from trade are near zero, a small deterioration in fundamentals can move the market from positive trade to none.²

¹Market unraveling after a small shock in the spirit of [Akerlof \(1970\)](#) is possible, but requires the demand curve and the average cost curve to be tangent—a knife-edge condition. See the Supplementary Material to [Azevedo and Gottlieb \(2017\)](#) for details.

²This knife-edge property is confirmed by Proposition 5 of [Attar et al. \(2011\)](#), which shows that whenever the good is traded in strictly positive quantity, equilibrium quantities remain positive after

Table 1: Conditions for Market Collapse from Small Cost Increases

Framework	Non-Excl. Contract.	Pooling in Equil.	Outside Opt. > Autarky	Cascade Possible?
Azevedo & Gottlieb (2017)	No	Varies	Yes	No*
Attar et al. (2011)	Yes	Full	No	Knife-edge*
Attar et al. (2014)	Yes	No	No	No
Attar et al. (2014) + Outside Option	Yes	No	Yes	No
Attar et al. (2021)	Yes	Partial	No	No
Attar et al. (2021) + Outside Option	Yes	Partial	Yes	YES

*Requires knife-edge conditions or large shocks.

Whether such unraveling is possible beyond this special case—with general preferences, non-linear pricing, and multi-layered equilibria—is the central question of our paper. We consider the two canonical generalizations: the simultaneous contracting environment of [Attar et al. \(2014\)](#) and the sequential contracting environment of [Attar et al. \(2021\)](#). We show that neither environment generates exit cascades from small cost increases alone. Once outside options are generalized, however, the environments diverge sharply: cascades become possible in [Attar et al. \(2021\)](#) but remain impossible in [Attar et al. \(2014\)](#). The difference, as we show, traces entirely to equilibrium structure—partial pooling in [Attar et al. \(2021\)](#) creates strategic complementarity in exit decisions that is absent from the separating equilibria of [Attar et al. \(2014\)](#). Table 1 summarizes the main message of this paper. Exit cascades outside of knife-edge cases are only possible when three features are met: non-exclusive contracting, (partial) pooling in equilibrium, and outside options exceeding the autarky payoff.

3 Model of [Attar et al. \(2021\)](#)

We present the model of [Attar et al. \(2021\)](#). The demand side of the market consists of a finite number of privately informed agents indexed by $i \in I \equiv \{1, \dots, n\}$ with a further exits. Complete unraveling therefore requires the market to be at the margin of viability to begin with.

strictly positive measure of each type, m_i . We normalize the total mass of agents to one, $\sum_{i \in I} m_i = 1$. Utility for each type is given by $u_i(q, t)$ and assumed to be continuous, strictly quasi-concave in the arguments³ and strictly decreasing in t . Here q represents the quantity of a good consumed and t is the transfer required to obtain it. An important feature of the model is that privately informed types are ordered by a single-crossing property (Milgrom and Shannon, 1994), $\forall i < j, q < q', t, t'$:

$$u_i(q, t) \leq u_i(q', t') \Rightarrow u_j(q, t) < u_j(q', t').$$

Single-crossing implies that a higher type is at least as willing as a lower type to trade an additional unit of the good for an additional transfer.⁴ Since utility functions need not be differentiable, we define the marginal rate of substitution without imposing differentiability. Let $\tau_i(q, t)$ be the supremum set of prices, p , such that

$$\tau_i(q, t) \equiv \sup \left\{ p : u_i(q, t) < \max_{q' \geq 0} u_i(q + q', t + pq') \right\}.$$

Hence, $\tau_i(q, t)$ is the marginal rate of substitution for agent i at consumption bundle (q, t) —the supremum price per unit at which agent i is willing to trade an additional quantity $q' > 0$ on top of (q, t) . An important assumption for our analysis is that, absent a transfer, a strictly positive endowment of q lowers agents' marginal rate of substitution:

Assumption 1 $\tau_i(q, 0) \leq \tau_i(0, 0), \forall i, q > 0$.⁵

By way of concrete examples, the model translates into the insurance economy

³Attar et al. (2021) maintain weak quasi-concavity and obtain an *essentially* unique entry-proof tariff: with weakly convex preferences, multiple optimal quantities arise when some type's marginal rate of substitution equals the layer price over a whole interval of quantities, a nongeneric phenomenon (see their discussion following Theorem 2). Because our comparative statics track the equilibrium across cost vectors, we impose the strict version, under which every layer (of the aggregate market tariff) problem has a unique solution and all uniqueness statements below are exact. For example, strict quasi-concavity holds in our leading insurance example. Throughout, we identify tariffs with their restriction to $[0, q_n]$.

⁴Here we are using the strict single-crossing condition rather than the weak version, which allows for equality. This is because we focus on the strict notion of market breakdown in light of Corollary 1 in Attar et al. (2021).

⁵The same assumption is also used in Attar et al. (2021) in their analysis.

of [Rothschild and Stiglitz \(1976\)](#), where i indicates an agent's probability of loss, q is the amount of insurance purchased, and t is the insurance premium. In a credit economy, i indexes borrower default probability, q is the loan quantity demanded, and t is the gross loan promise made to the lender.

Contracts, defined by the pair (q, t) for $q \geq 0$, are supplied by competitive risk-neutral, expected profit maximizers. The suppliers possess a linear production technology with a unit cost, $c_i > 0$, which is increasing in i . Adverse selection arises because c_i is increasing in type: higher types wish to trade more but are more costly to serve. Define by $\bar{c}_i$ the expected unit cost of serving $j \geq i$ (the upper-tail conditional expected cost) given that higher types will be willing to trade any contract offered to type i .⁶ Formally,

$$\bar{c}_i \equiv E[c_j | j \geq i] = \frac{\sum_{j \geq i} m_j c_j}{\sum_{j \geq i} m_j}. \quad (1)$$

3.1 Equilibrium

We start with an equilibrium in which the market is inactive: there is no set of contracts an entrant can propose that leads to positive trades with nonnegative profits. Such a market is “entry-proof” in the sense of [Attar et al. \(2021\)](#).

The key is to first consider the no-trade contract, $(0, 0)$, as any agent's outside option. (Our subsequent analysis generalizes the outside option utility to be greater than that of the no-trade contract.) Then, a market is entry proof iff, for any menu of contracts an entrant offers, the buyer's best response earns the entrant zero expected profit. That is, the entry-proof condition is given by

$$\textbf{Condition EP :} \quad \tau_i(0, 0) \leq \bar{c}_i \quad \forall i. \quad (2)$$

Condition EP simply says that there will be no trade in a market when the cost of offering a contract for agent i exceeds type i 's marginal utility of not trading,

⁶Since c_i is strictly increasing in i and each $m_i > 0$, the upper-tail conditional expected cost $\bar{c}_i$ is strictly increasing in i . This is what makes the layered market tariff below convex.

given that all other types higher than i must also be served.⁷ Theorem 1 in [Attar et al. \(2021\)](#) states that **Condition EP** is necessary and sufficient for markets to be inactive.

Now we consider an equilibrium in which the market is active, so there is positive quantity of trades. Trade in the market is non-exclusive in the sense that no agent can be stricken from trading with multiple firms or suppliers. Therefore, we must define the market tariff, or the minimum aggregate transfer that is made across active markets to obtain aggregate consumption, q . Define the aggregate market tariff by $T(q)$. We will assume that $T(q)$ is convex and the domain is a compact interval with lower bound equal to 0.⁸ Then, all agents choose q_i to maximize $u_i(q_i, T(q_i))$. An allocation, $(q_i, T(q_i))_{i \in I}$ is *implemented* by the market tariff, T , if $q_i = \arg \max_q u_i(q, T(q))$. This allocation is *budget feasible* if suppliers make non-negative expected profits at the market tariff

$$\sum_i m_i [T(q_i) - c_i q_i] \geq 0. \quad (3)$$

Under an exclusive contracting environment, an entrant only needs to consider the agent's direct utility functions. However, under a non-exclusive contracting environment, an agent can combine any menu of potential new contracts with trades along the existing market tariff, T . Thus, an active market is entry proof if an entrant cannot make positive expected profits given agents' ability to combine contracts. Therefore, the entrant faces types with indirect utility functions of trading a proposed new contract (q', t') in addition to the existing allocation $(q, T(q))$:

$$u_i^T(q', t') \equiv \max \{u_i(q + q', T(q) + t') : q \geq 0\}, \quad (4)$$

From the entrant's perspective, an agent's individual rationality constraint is determined by the indirect utility of not trading the proposed contract on top of the

⁷We use contracts and markets interchangeably, treating all contracts with potentially different price-quantity pairs as individual markets following [Bisin and Gottardi \(1999\)](#); [Dubey and Geanakoplos \(2002\)](#); [Dubey et al. \(2005\)](#).

⁸See the discussion at the end of this section for more details on strategic foundations of $T(q)$.

existing market tariff, $u_i^T(0, 0)$. We can define the marginal rate of substitution along the indirect utility functions as above by $\tau_i^T(q', t')$.⁹ As shown by Attar et al. (2021), the indirect utility functions also satisfy single-crossing, since the primitives do.

Attar et al. (2021) impose an additional assumption, slightly stronger than Assumption 1, requiring that the marginal rate of substitution is nonincreasing in q for any type i and transfer t :

Assumption 2 *For all i and t , $\tau_i(q, t)$ is nonincreasing in q .*

Intuitively, this assumption implies that a higher quantity always reduces each agent's willingness to pay for any additional quantity.¹⁰ For the given market tariff, T , and the allocation, $(q_i, T(q_i))$, we can define $\tau_i^T(0, 0)$ as the supremum of the set of prices p for the indirect utility function, $u_i^T(0, 0)$ —that is,

$$u_i(q_i, T(q_i)) = u_i^T(0, 0) < \max \{u_i^T(q', pq') : q'\} = \max \{u_i(q + q', T(q) + pq' : q, q')\}.$$

With this, we can invoke the necessity and sufficiency result of **Condition EP** to state that a market tariff is entry-proof iff:

$$\forall i, \quad \tau_i^T(0, 0) \leq \bar{c}_i. \quad (5)$$

This condition states that an active market is entry-proof if and only if the cost required to enter the market exceeds the willingness of each agent to trade the contract on top of the allocation they may already obtain. For each agent, the utility they receive from their market trades must be at least as large as trading the proposed

⁹This is possible because: (i) the maximizers in (4) are continuous by Berge's Maximization Theorem; (ii) the market tariff T is convex; (iii) the utility functions $u_i(q, t)$ are weakly quasi-concave in (q, t) and strictly decreasing in t ; and (iv) therefore the indirect utility functions u_i^T are weakly quasi-concave in (q, t) and strictly decreasing in t .

¹⁰Throughout the paper, we maintain the demand-finiteness condition of Attar et al. (2021) in the following locally uniform form: for every compact set of bundles (q, t) and every $\underline{p} > 0$, the set $\bigcup \arg \max \{u_i(q + q', t + pq') : q' \geq 0\}$, where the union is over i , bundles in the set, and prices $p \geq \underline{p}$, is nonempty and bounded. This guarantees that every layer problem below admits a solution and that optimal quantities are bounded uniformly over the relevant neighborhoods of cost vectors.

new contract given the cost required to serve the market:

$$\text{for each } i, \quad u_i(q_i, T(q_i)) \geq \max \{u_i(q + q', T(q) + \bar{c}_i q' : q, q')\}. \quad (6)$$

From the entrant's perspective, by convention of letting $q_0 \equiv 0$, setting $q' = q_i - q$, and applying condition (6) for each layer, $q \in [q_{i-1}, q_i]$, we see that the implied market tariff necessary to induce entry must be at least as large as the pre-entry tariff:

$$\text{for each } i \text{ and } q \in [q_{i-1}, q_i], \quad T(q_i) \leq T(q) + \bar{c}_i(q_i - q). \quad (7)$$

For a given $q = q_{i-1}$, we have $T(q_i) \leq T(q_{i-1}) + \bar{c}_i(q_i - q_{i-1})$. Using the budget-feasibility condition from (3), and re-writing it in terms of layers, we have

$$\sum_i \left(\sum_{j \geq i} m_j \right) [T(q_i) - T(q_{i-1}) - \bar{c}_i(q_i - q_{i-1})] \geq 0. \quad (8)$$

Therefore, combining (7) and (8) implies $T(q_i) = T(q_{i-1}) + \bar{c}_i(q_i - q_{i-1})$. It must also be true that the allocation, $u_i(q_i, T(q_i))$, implied by the new layer, $q_i - q_{i-1}$ maximizes the utility of all the agents that choose it, given that the new market tariff must rise to serve all agents. Hence, for each i ,

$$u_i(q_i, T(q_i)) = \max \{u_i(q_{i-1} + q', T(q_{i-1}) + \bar{c}_i q' : q')\}. \quad (9)$$

Finally, because the tariff is convex and satisfies (7) and (9), it must be affine with slope $\bar{c}_i$ over the interval $[q_{i-1}, q_i]$.

With this, [Attar et al. \(2021\)](#) state **Theorem 2**—An allocation $(q_i, T(q_i))_{i \in I}$ is budget-feasible and implemented by an entry-proof convex market tariff, T , with domain $[0, q_n]$ if and only if

1. $(q_0, T(q_0)) \equiv (0, 0)$,
2. $q_i - q_{i-1} \in \arg \max \{u_i(q_{i-1} + q', T(q_{i-1}) + \bar{c}_i q' : q')\}$ for each i ,
3. $q_{i-1} < q_i \Rightarrow T$ is affine with slope $\bar{c}_i$ over $[q_{i-1}, q_i]$ for each i .

To sum up, an affine convex market tariff is entry proof as long as the upper-tail conditional cost of entry exceeds the marginal willingness of all agents to trade the additional layer on top of their pre-entry allocation. This market tariff consists of layers of trade with unit prices $\bar{c}_i$ that trace out a polygon with an upward kink at each $q_{i+1} \geq q_i$ for each $i \in I$.

We take the strategic foundations of the allocations characterized in **Theorem 2** above as given. There are two ways to micro-found this allocation. The first is the discriminatory ascending-price auction used in [Attar et al. \(2021\)](#). In particular, each time a new price is quoted, each seller publicly announces the maximum quantity he stands ready to trade at this price. Once this auctioning phase is completed, the buyer decides which quantities to purchase from which sellers in a non-exclusive way. The second is the notion of competitive-pooling in general equilibrium pioneered by [Dubey and Geanakoplos \(2002\)](#) and extended to a non-exclusive contracting by [Dubey and Geanakoplos \(2019\)](#).

4 Cost Increases and Market Shutdowns

We now characterize how the equilibrium allocation responds when the cost of serving the market increases. Such cost increases may reflect a deterioration in the quality of the agent pool—for example, a surge in borrower defaults—or any other shock that raises the unit cost c_i . We define a *market shutdown* as the comparative static in which a previously active market becomes inactive: trade falls to zero.

To fix ideas, recall that the unit cost to supply a contract for each type is c_i . Now suppose that there is a cost shock by which the cost to serve types increase to $\tilde{c}_i = c_i + \epsilon_i$, $\epsilon_i \geq 0$, $\forall i \in I$. The resulting upper-tail conditional expected cost of serving all agents must rise as long as there exists at least some type with $\epsilon_i > 0$. Since we are interested in small shocks, we assume that the order of cost is maintained after the cost increase, i.e., $\tilde{c}_i < \tilde{c}_j$, for all $i < j$. Define the new upper-tail conditional

expected cost for any agent i by

$$\tilde{\bar{c}}_i \equiv \frac{\sum_{j \geq i} m_j \tilde{c}_j}{\sum_{j \geq i} m_j}, \quad (10)$$

and $\tilde{\bar{c}}_i \geq \bar{c}_i$. Then, it is clear that any market that was inactive *ex ante* the cost increase remains inactive *ex post* because **Condition EP** implies $\tau_i(0, 0) \leq \bar{c}_i \leq \tilde{\bar{c}}_i$. Intuitively, making the average quality of the pool worse will never lead to the opening of new markets.

When do cost increases lead any active markets to become inactive and shut down? An active market satisfies $\tau_i(0, 0) > \bar{c}_i$ for some i . The cost increase from c_i to $\tilde{c}_i$ raises the slope of the market tariff weakly along every segment of the polygon tracing out the new market tariff, $\tilde{T}(\tilde{q})$, so each remaining type faces weakly worse terms of trade. Nevertheless, participation is governed by the marginal comparisons of **Condition EP**, and complete shutdown requires $\tau_i(0, 0) \leq \tilde{\bar{c}}_i$ for *every* i , which by (10) requires sufficiently large cost increases. The theorem below makes both statements precise and adds a local robustness result: when the pre-shock equilibrium is active with slack, no active layer disappears after a sufficiently small shock.

Formally, a layer i is *active* at the equilibrium allocation if $q_i > q_{i-1}$. By the definition of the marginal rate of substitution, an active layer satisfies the participation inequality

$$\tau_i(q_{i-1}, T(q_{i-1})) \geq \bar{c}_i, \quad (11)$$

and strict inequality in (11) guarantees activity. We say the pre-shock equilibrium is *active with slack* if (11) holds with strict inequality for every active layer. This fails only when a type is exactly indifferent to its layer.

Theorem 1 (Cost Increases and Market Shutdowns)

Suppose c_i weakly increases to $\tilde{c}_i = c_i + \epsilon_i$ for all $i \in I$, where $c_j < \tilde{c}_j$ for some $j \in I$ and $\tilde{c}_1 < \dots < \tilde{c}_n$. Then, for any new equilibrium affine aggregate tariff $\tilde{T}$ and new quantities $(\tilde{q}_i)_{i \in I}$, the following hold:

- (i) *All inactive markets (market shutdown) remain inactive under the new equilib-*

rium.

- (ii) *Each type's layer carries a weakly higher unit price: $\tilde{T}$ is affine with slope $\tilde{c}_i \geq \bar{c}_i$ over $[\tilde{q}_{i-1}, \tilde{q}_i]$ for every i with $\tilde{q}_i > \tilde{q}_{i-1}$.*
- (iii) *If the pre-shock equilibrium is active with slack, there exists $\bar{\epsilon} > 0$ such that, whenever $\max_{i \in I} \epsilon_i < \bar{\epsilon}$, every pre-shock active layer remains active in every post-shock equilibrium.*
- (iv) *Complete market shutdown ($\tilde{q}_i = 0$ for every $i \in I$) occurs if and only if $\tau_i(0, 0) \leq \tilde{c}_i$ for every $i \in I$.*

Theorem 1 states general properties of active and inactive markets under the new equilibrium. One important implication is that active markets will completely shut down only if $\tau_i(0, 0) \leq \tilde{c}_i$ for all i . Hence, the cost increase must be sufficiently large to generate market shutdowns.¹¹ Our market shutdown result extends the Akerlof-like unraveling result established in the non-exclusive contracting setting of Attar et al. (2011) to more general preferences and non-linear pricing. Their paper shows that even in the strategic setting of Rothschild and Stiglitz (1976), equilibrium always exists and is unique with non-exclusive competition. Hence markets fail in the Akerlof sense when the unique equilibrium is the no-trade equilibrium.

Moreover, Theorem 1 implies that no trade occurs in non-exclusive settings even when the separating equilibrium of Rothschild and Stiglitz (1976) exists under an exclusive contracting environment. For example, consider the 2-agent case. Under the exclusive contracting environment a pure-strategy equilibrium exists as a separating equilibrium as long as the payoff from a separating contract dominates the payoff from a pooling contract as

$$u_1(q_{RS}, c_1 q_{RS}) \geq \max_q u_1(q, \bar{c}_1 q),$$

¹¹Azevedo and Gottlieb (2017) establish a similar result in the exclusive contracting environment. We extend their finding to the non-exclusive contracting case.

where q_{RS} is the quantity that satisfies

$$\max_q u_2(q, c_2 q) = u_2(q_{RS}, c_1 q_{RS}).$$

Under this simplified setup, the market shutdown condition under the non-exclusive contracting environment is

$$\begin{aligned} \tau_1(0, 0) &\leq \bar{c}_1, \quad \& \quad \tau_2(0, 0) \leq c_2 \\ \Rightarrow u_1(0, 0) &\geq \max_q u_1(q, \bar{c}_1 q), \quad \& \quad u_2(0, 0) \geq \max_q u_2(q, c_2 q). \end{aligned}$$

Combining the equilibrium existence condition for [Rothschild and Stiglitz \(1976\)](#) and our entry-proof market shutdown condition implies

$$u_1(q_{RS}, c_1 q_{RS}) \geq u_1(0, 0) \geq \max_q u_1(q, \bar{c}_1 q),$$

hence, introducing non-exclusive contracting excludes the best (lowest-cost) type from trade for the same parameter values that guarantee existence of a separating equilibrium under exclusive contracting. Complete shutdown additionally requires $\tau_2(0, 0) \leq c_2$, which under expected utility is possible only if the worst type incurs a loss with certainty (cf. [Attar et al., 2021](#), p. 2633). The intuition is as follows. Under non-exclusive contracting, full separation is not attainable and equilibrium always contains partial pooling. Trade for agent 1 shuts down when the no-trade contract dominates the partial pooling contract for agent 1, which occurs when the mass of bad types is sufficiently large. Under exclusive contracting, by contrast, the separating equilibrium exists precisely when the mass of bad types is large enough to make the pooling contract unattractive relative to full separation. Since no trade under non-exclusive contracting requires a larger mass of bad types than what makes pooling unattractive under exclusive contracting, the exclusion of agent 1 under non-exclusive contracting implies existence of a separating equilibrium under exclusive contracting.

Our next result addresses which layers are most susceptible to agents exiting. The single-crossing condition implies that higher types have higher willingness to trade at

higher prices. Hence, as costs increase the market tariff along each layer, the lowest types exit the market first. This is easiest to see for the case where the first type $i = 1$ is just indifferent to trading given the upper-tail conditional cost of serving all greater types $j > 1$ where $\tau_i(0, 0) = \bar{c}_i < \tilde{c}_i$, while other agents $j > 1$ could still have $\tau_j^{\tilde{T}}(0, 0) > \tilde{c}_j$. In this case, type 1 is no longer willing to trade at the new market tariff, $\tilde{T}(\tilde{q})$, given the additional cost required to make the tariff budget-feasible while all types $j > 1$ remain active in the market. Hence, only the first layer of trade, q_1 , becomes inactive.

Proposition 1 *In equilibrium, the lowest type (cost) agents exit first as costs increase, and the active set is an upper interval.*

An important implication of Proposition 1 is that *small* cost increases have only marginal negative effects on active markets. The reason is that, absent outside options, participation is governed by the marginal conditions of **Condition EP**. The remaining market shuts down only if $\tau_i(0, 0) \leq \tilde{c}_i$ for every $i \geq 2$, and neither side of this inequality is affected by type 1's exit. Remaining agents' total utility falls, as they lose the cheapest layer, but the marginal incentives that keep markets open are preserved. This force of marginal incentives keeps the remaining markets active. Therefore, **Condition EP** is robust in the sense that complete market unraveling under non-exclusive contracting requires a sufficiently large cost increase, consistent with [Azevedo and Gottlieb \(2017\)](#).

5 Outside Options and Market Shutdown

We now show that a cascade of exits—a sequential shutdown of multiple markets—is possible upon small increases in costs when outside options are present. An outside option represents an alternative to market participation: it may reflect a contractual barrier, a cost of entry such as the time and effort required to sign contracts or search for suppliers, or access to an external market that offers a reservation utility. The key feature is that agents compare the value of their outside option to the utility they obtain from market trades.

Denote the utility from outside options (and not entering the market) as γ_i for type i . Type i now compares the set of market contracts to their outside option. Thus, the individual rationality condition for i becomes

$$V_i(T) = \max \left\{ \gamma_i, \max_{q \geq 0} u_i(q, T(q)) \right\} \quad (12)$$

for a given aggregate market tariff T .¹²

The extensive form of the game with generalized outside options is the following:

1. After privately learning her type, each buyer chooses whether to take the outside option or enter the market, given her rational expectations about the market equilibrium.
2. Given the composition of entrants, suppliers and entrants play the game of [Attar et al. \(2021\)](#) and the resulting equilibrium determines the entry-proof aggregate market tariff.

Participation decisions are not observed by suppliers, who post tariffs against the anticipated composition of entrants. A buyer who deviates at the participation stage therefore faces the tariff that the market posts for the anticipated participant set, and, in particular, a buyer who enters a market that no other type enters finds no contracts available and obtains the autarky payoff, $u_i(0, 0)$.

We assume a tie-breaking rule whereby an agent who is indifferent between market trade and the outside option exits the market (i.e., if $\max_{q \geq 0} u_i(q, T(q)) = \gamma_i$, type i exits the market). This tie-breaking assumption is made for simpler exposition, as our main results remain unchanged if it is altered.

Because participation precedes contracting, the two stages can be analyzed in turn. Fix a candidate participant set. The continuation game is exactly the environment of [Attar et al. \(2021\)](#) for the truncated type distribution: suppliers compete for a fixed set of buyers, and the characterization of budget-feasible, entry-proof convex tariffs applies to it in full, against arbitrary menus of contracts. It remains to ask which participant sets are consistent with the participation decisions that generate them.

¹²Note outside options are irrelevant for equilibrium when they provide utility less than no trade.

For a candidate active set, let T^k denote the entry-proof layered tariff when types below k do not participate, and define $U_j^k \equiv \max_{q \geq 0} u_j(q, T^k(q))$. For notational completeness, let T^{n+1} denote the degenerate no-trade tariff with domain $\{0\}$ and $T^{n+1} = 0$.

The formal definition of our equilibrium concept is the following:

Definition 1 *An equilibrium consists of an active set $I_A \subseteq I$ and an aggregate market tariff T such that:*

- (i) *given T , each type i participates if and only if $\max_{q \geq 0} u_i(q, T(q)) > \gamma_i$, and I_A is the set of participating types;*
- (ii) *given I_A , T is a budget-feasible, entry-proof tariff of the continuation game of [Attar et al. \(2021\)](#) played among the participants I_A ; and*
- (iii) (No self-fulfilling exits.) *every non-participating type $j \notin I_A$ satisfies $U_j^j \leq \gamma_j$.*

Clause (iii) is a participation refinement or equilibrium-selection refinement. Clauses (i) and (ii) alone can support self-fulfilling low-participation configurations because a type may prefer to participate when it is expected to participate, but prefer to remain out after its own absence has worsened the anticipated tariff. Clause (iii) rules out exactly such self-fulfilling exits: an excluded type must prefer its outside option even to the market in which it would be the best remaining participant. Proposition 4 shows that this criterion selects the participation-consistent configuration with the largest active set.¹³

5.1 Cascade of Exits

In this subsection, we present our main result: a sequential market shutdown arising from a cascade of exits, in which the exit of one agent triggers another's. The mechanism rests on the spillover effects of exits across agents in the presence of outside

¹³Two considerations support this selection. Buyers prefer it: by Lemma 2, the type immediately above the marginal participant is strictly better off when that participant remains, and the same force operates on every higher type. And it is what entry would enforce if participation were revisable after contracts are posted: when $U_j^j > \gamma_j$, an entrant offering a capped linear menu priced slightly above $\bar{c}_j$ attracts type j and, by (13), the entire upper tail, at a strictly positive profit.

options. These spillover effects are amplified through partial pooling, because each agent's exit generates a lump-sum decline in utility for all other agents participating in the same pool. If the initial exit triggers subsequent exits, the impact of each additional exit accumulates, generating amplification through multiple exits. We show that the spillovers needed to generate a market shutdown do not exist in the model without outside options.

We assume the following condition on the distribution of outside option values:

$$u_i(q, t) \geq \gamma_i \Rightarrow u_j(q, t) > \gamma_j \quad \forall i < j, \forall q > 0, \forall t. \quad (13)$$

This condition implies that if agent i prefers a contract (q, t) with a positive quantity to the outside option, then agent $j > i$ also prefers the contract to the outside option, which is in line with the idea of single-crossing property. We discuss the role and micro-foundation of this assumption in Appendix B.

Once the set of entrants is fixed, the continuation contracting problem is the same non-exclusive entry-proofness problem as in the baseline model, so Theorem 2 of Attar et al. (2021) applies directly to the truncated subeconomy. The remaining issue is therefore not stage-2 re-attraction, but stage-1 participation consistency and equilibrium selection. Proposition 3 establishes both.

The aggregate entry-proof convex market tariff T has slope $\bar{c}_i$ over each layer $[q_{i-1}, q_i]$, with the convention $q_0 = 0$. Accordingly, Statement (i) of Theorem 1, the tariff-slope monotonicity in its Statement (ii), and Proposition 1 continue to hold for any fixed set of entrants. And participation itself is now governed by the γ_i comparisons below. The small-shock robustness in Statement (iii) of Theorem 1 also extends whenever the participation constraints $\max_{q \geq 0} u_i(q, T(q)) > \gamma_i$ hold with strict slack for all active types. The exit cascades of this section arise precisely when this slack vanishes for the marginal type. In particular, the exit ordering of Proposition 1 yields the following cutoff characterization.

Proposition 2 *In any active market equilibrium, there is a cutoff-type $\theta \in I \cup \{0\}$ such that any agents with type less than or equal to θ exit the market and any agents*

with type greater than θ remain in the market.

Proposition 2 follows the same intuition of Proposition 1, as the lowest type agents exit first in any active market equilibrium. Proposition 2 is important for eliminating any possible nonexistence of equilibrium (details are discussed in Supplementary Appendix C.2).

Given the outside option values $\{\gamma_i\}$, satisfying (13), the active set is an upper interval $I_A = \{\theta + 1, \theta + 2, \dots, n\}$, and the restricted market tariff only accounting for the set I_A defined as $T^{\theta+1}$ is the unique budget-feasible, entry-proof convex aggregate tariff for the entrants I_A . It remains to check that the participation decisions generating I_A are consistent: the participants prefer the market to their outside options, and each exited type prefers its outside option both to the market it faces and to the market in which it would be the best remaining participant. The following proposition establishes both.

Proposition 3 *Suppose $\{\gamma_i\}_{i \in I}$ satisfy (13) and $\gamma_i \geq u_i(0, 0)$ for every i . Fix a candidate active set equal to the upper interval $I_A = \{\theta + 1, \dots, n\}$ (Proposition 2), and let $T^{\theta+1}$ be the Attar et al. (2021) layered tariff for the subeconomy I_A : the convex tariff that is affine with slope $\bar{c}_i = \mathbb{E}[c_j \mid j \geq i]$ on each layer $[q_{i-1}, q_i]$, $i \in I_A$, with $q_\theta = 0$. Then:*

- (i) (The characterization carries over.) $T^{\theta+1}$ is the unique budget-feasible, entry-proof aggregate tariff for the entrants I_A .
- (ii) (Consistency of the participant set.) Suppose $U_i^i \leq \gamma_i$ for every $i \leq \theta$ and $U_j^{\theta+1} > \gamma_j$ for every $j \in I_A$. Then $(I_A, T^{\theta+1})$ is an equilibrium in the sense of Definition 1. Conversely, in any equilibrium with active set I_A , every exited type $i \leq \theta$ satisfies $U_i^i \leq \gamma_i$.

Proposition 3 shows that the equilibrium concept of the baseline model carries over to the model with generalized outside option extension when the outside option values also have single-crossing property. Statement (ii) of the proposition shows that, with generalized outside options, the exit margin is governed by $U_i^i \leq \gamma_i$ rather

than by the baseline condition $\tau_i(0, 0) \leq \bar{c}_i$. An exited type may satisfy $\tau_i(0, 0) > \bar{c}_i$, so that suppliers would gladly serve it, yet the type may stay out because $U_i^i \leq \gamma_i$. It is exactly this wedge that makes cascades possible: a cost increase that pushes U_i^i below γ_i removes a type the baseline model would have kept active.

Given that the participation set is an upper interval, the equilibrium active set can be computed constructively. We define the process by which exits unfold and then show that it selects the maximal-participation equilibrium.

Definition 2 (Sequential exit process) *Fix a cost vector $\tilde{c}$. Set $A_0 \equiv I$. At step $t \geq 1$, given an active set $A_{t-1} = \{k, \dots, n\}$, the lowest active type k exits if and only if $U_k^k \leq \gamma_k$, in which case $A_t = \{k+1, \dots, n\}$. The process stops the first time the lowest active type strictly prefers market participation, or when the active set is empty. Denote the terminal active set by A^* .*

At each step the exit test is exactly the participation comparison of Definition 1(i), under the tie-breaking rule, facing the tariff for the current active set. The process terminates after at most n steps.

The sequential exit process of Definition 2 is the constructive counterpart of Definition 1: it begins with all types active and removes the lowest active type whenever that type prefers its outside option to the market in which it is the best remaining participant, terminating at $A^* = \{k^*, \dots, n\}$ with $k^* = \min\{k : U_k^k > \gamma_k\}$.

Proposition 4 (Maximal participation) *The terminal set A^* of the sequential exit process contains the active set of every configuration satisfying clauses (i) and (ii) of Definition 1. Among participation-consistent configurations, clause (iii) selects the one with the largest set of participants.*

Without clause (iii), participation decisions are strategic complements and several configurations can satisfy clauses (i) and (ii). If

$$U_k^{k+1} \leq \gamma_k < U_k^k,$$

type k participates when it is expected to participate (and thereby generates the tariff T^k under which participation is optimal), but prefers to stay out when it is expected to stay out and the market posts T^{k+1} . Both configurations satisfy clauses (i) and (ii). Only the first satisfies clause (iii). Supplementary Appendix C.2 discusses this complementarity, in the sense of Bulow et al. (1985), in more detail.

Clause (iii) selects the maximal-participation configuration by ruling out an exit that is sustainable only because the exiting type's own absence has already worsened the market. We adopt this refinement throughout the analysis. No separate stage-2 re-entry argument is needed because the participation decision is made before the continuation contracting game.

We now use Proposition 2 to discuss how exits of the lowest types can generate a further cascade. Agents decide to enter or remain in the market by comparing their total utility from market trades with the value of their outside option, so they exit whenever the outside option is preferable. Because the lowest types exit first, all remaining agents face a higher aggregate market tariff, which lowers their total utility. Unlike the baseline model without outside options—where remaining agents only need their marginal rate of substitution at no-trade to exceed the market cost, as in (2)—agents with outside options compare their *total utility* from market trades against γ_i . This comparison can trigger a cascade: even agents who would have stayed under the baseline may exit once their total utility falls below their outside option threshold.

We now formalize the cascade argument.¹⁴ Recall that T^k denotes the entry-proof layered tariff when types below k do not participate,¹⁵ and that the maximized utility of type j under this tariff is $U_j^k \equiv \max_{q \geq 0} u_j(q, T^k(q))$. Explicitly, by Proposition 3(i) applied to the upper interval $\{k, \dots, n\}$, on each layer $q \in [q_{\ell-1}^k, q_\ell^k]$ with $\ell \geq k$ the tariff is

$$T^k(q) = \sum_{s=k}^{\ell-1} \tilde{c}_s(q_s^k - q_{s-1}^k) + \tilde{c}_\ell(q - q_{\ell-1}^k), \quad (14)$$

¹⁴Supplementary Appendix C.1 provides a detailed illustration of cascade of exits.

¹⁵By Proposition 2, all types above k participate whenever type k participates, so $\{k, \dots, n\}$ is the relevant active set.

with the convention $q_{k-1}^k = 0$, where $q_\ell^k - q_{\ell-1}^k \in \arg \max_{q \geq 0} u_\ell(q_{\ell-1}^k + q, T^k(q_{\ell-1}^k) + \tilde{c}_\ell q)$. The tariff is continuous and convex, with slope $\tilde{c}_\ell$ on each layer, and all objects are evaluated at the post-shock costs $\tilde{c}$.

Exit decisions are sequential—one type’s exit changes the tariff the next type faces, and our theorem must distinguish exits that are *triggered* by a preceding exit from exits that would have occurred anyway. The sequential exit process of Definition 2 describes how exits unfold. The following definition isolates the cascade property of an equilibrium.

Definition 3 (Exit cascade) *Fix $2 \leq i \leq n + 1$. An equilibrium (I_A, T) in the sense of Definition 1 exhibits an exit cascade up to type i if the following hold:*

- (a) $I_A = \{i, \dots, n\}$ (with $I_A = \emptyset$ when $i = n + 1$); and
- (b) *every exit after the first is triggered by its predecessor’s: for each $j \in \{2, \dots, i - 1\}$, type j would strictly prefer trading along the tariff T^{j-1} (the market that would prevail were type $j - 1$ retained together with all higher types) to its outside option; equivalently, $\gamma_j < U_j^{j-1}$.*

Clause (b) separates a genuine cascade from a collection of independent exits: if $\gamma_j \geq U_j^{j-1}$, type j would have exited even had type $j - 1$ remained in the market, so its exit is caused by the cost shock directly rather than by the preceding exit.

The proof of the cascade theorem repeatedly compares a type’s market utility across active sets. The following lemma provides the two facts we need. In particular, statement (ii) shows that once a type weakly prefers its outside option to the market in which it is the *best* remaining participant, it also weakly prefers the outside option to any market from which additional lower types have been removed.

Lemma 1 (Layer-price bounds) *For every $k \in I$ and every type $j \leq k$:*

- (i) $U_k^k = \max_{q \geq 0} u_k(q, \tilde{c}_k q)$;
- (ii) $U_j^k \leq \max_{q \geq 0} u_j(q, \tilde{c}_k q) \leq U_j^j$.

In particular, if $U_j^j \leq \gamma_j$, then $U_j^k \leq \gamma_j$ for every $k \geq j$.

Lemma 1 governs the types that have already exited. Once a type weakly prefers its outside option to the market in which it is the best remaining participant, removing additional lower types cannot restore its participation incentive under the surviving tariff. The next lemma establishes the complementary (and, for the cascade, decisive) fact: each remaining type is strictly harmed by its predecessor's exit. It is the negative participation externality that outside options convert into exits.

Lemma 2 (Predecessor-exit spillover) *Fix $1 < j \leq n$ and post-shock costs. Then $U_j^{j-1} \geq U_j^j$. The inequality is strict whenever type $j-1$'s layer under T^{j-1} has positive length, $q_{j-1}^{j-1} > 0$, and type j trades a positive quantity at the price $\tilde{c}_j$, that is, $q_j^j > 0$.*

Suppose that agent 1 exits the market after costs increase, which occurs when **Condition IT** holds:

$$U_1^1 \leq \gamma_1. \quad (15)$$

Note that we do not need to check **Condition EP**, $\tau_1(0, 0) \leq \tilde{c}_1$, because each agent weakly prefers the outside option over the no-trade contract, $\gamma_j \geq u_j(0, 0)$ for all j . A cascade requires more of each subsequent exit: type j must prefer to participate when $j-1$ participates, yet prefer to exit once $j-1$ has exited. We denote this two-sided requirement **Condition MS**(i):

$$U_j^j \leq \gamma_j < U_j^{j-1}, \quad \forall j < i \text{ and } j \neq 1. \quad (16)$$

By Lemma 2, $U_j^j < U_j^{j-1}$ under the stated conditions, so for each j , the set of outside options satisfying **Condition MS**(i) is a nonempty interval.

Finally, for the cascade to stop at type i (rather than continue further), the cutoff type must strictly prefer the market in which it is the best remaining participant. We denote this **Condition ST**(i):

$$\gamma_i < U_i^i, \quad (17)$$

with the convention that **Condition** $\mathbf{ST}(n+1)$ holds vacuously (the case of complete market shutdown). **Condition** $\mathbf{MS}(i)$ differs from **Condition** $\mathbf{EP}$ in two ways: the exit margin is *total* utility against the outside option rather than the marginal rate of substitution at no trade, and the right inequality additionally requires that each exit be triggered by the preceding one. Agents who would remain active under **Condition** $\mathbf{EP}$ may nonetheless exit when their total utility from trade falls below their outside option. We now show that these conditions are necessary and sufficient for an exit cascade.

Theorem 2 (Cascade of Exits) *For $2 \leq i \leq n+1$, suppose that (13) holds, $\gamma_j \geq u_j(0,0)$ for all $j \in I$, and post-shock costs are $\tilde{c}$ with $\tilde{c}_1 < \dots < \tilde{c}_n$. Then the post-shock economy has an equilibrium (in the sense of Definition 1) that exhibits an exit cascade up to type i (types $1, \dots, i-1$ exit, each exit after the first triggered by its predecessor's, and types $i, \dots, n$ remain active) if and only if **Conditions** $\mathbf{IT}$, $\mathbf{MS}(i)$, and $\mathbf{ST}(i)$ hold.*

Two remarks on the proof. First, U_j^j is the benchmark participation value: it asks whether type j would prefer market participation when it is treated as the best remaining participant and faces the break-even upper-tail conditional expected cost as tariff. Second, the proof uses condition (13) directly only in sufficiency, to propagate the cutoff type's participation to all higher types. The remaining steps rely on the layer-price bound of Lemma 1.

An immediate byproduct is that condition (13) restores the existence and uniqueness properties of the baseline model (under the maximal-participation refinement), formalizing the discussion following Proposition 2:

Corollary 1 (Existence and uniqueness) *Suppose the outside options satisfy (13) and $\gamma_j \geq u_j(0,0)$ for all $j \in I$. Then, given the tie-breaking rule, an equilibrium in the sense of Definition 1 exists, its active set is the terminal set A^* of the sequential exit process (Definition 2), and the equilibrium outcome (the active set and the market tariff) is unique.¹⁶*

¹⁶Absent clause (iii) of Definition 1, configurations satisfying clauses (i) and (ii) alone need not be unique, and A^* is the largest of them (Proposition 4).

Corollary 2 (Small shocks, large cascades) Fix $2 \leq i \leq n + 1$ and pre-shock costs c . Suppose the outside options satisfy, at the pre-shock costs,

$$\gamma_1 < U_1^1(c), \quad U_j^j(c) < \gamma_j < U_j^{j-1}(c) \quad \text{for every } 1 < j < i, \text{ and } \gamma_i < U_i^i(c).$$

Then, there exists $\bar{\epsilon} > 0$ such that every cost increase with $\max_k(\tilde{c}_k - c_k) < \bar{\epsilon}$ and $U_1^1(\tilde{c}) \leq \gamma_1$ produces an equilibrium exhibiting an exit cascade up to type i . Only the triggering type must be near its participation threshold, and every subsequent exit occurs with strict slack in its direct shock threshold and is caused by its predecessor's exit.

Some threshold proximity is unavoidable, as if every type carries a large participation surplus, no small perturbation moves the market. However, the corollary highlights that proximity is needed only for the trigger, therefore, the propagation itself is not a knife-edge case.

Setting $i = n + 1$ in Theorem 2 (so that **Condition ST**($n + 1$) is vacuous), **Conditions IT** and **MS**($n + 1$) characterize complete market shutdown through a cascade: every type exits, and by Corollary 1 the no-trade outcome is the unique equilibrium satisfying 1. A cost increase just large enough to trigger **Condition IT** for the best type can therefore shut down the entire market, provided the remaining inequalities in **MS**($n + 1$) hold.

Outside options have two roles. First, they generate discontinuous jumps in market quantities, q_i , after small cost increases. A small cost increase will trigger type i agent to exit with strictly positive quantity q_i when i 's utility is close to the utility from the outside option as $u_i(q_i, T(q_i)) \approx \gamma_i$. By contrast, without outside options, quantities vary continuously with costs at regular equilibria, and no active layer disappears under sufficiently small shocks (Theorem 1(iii)). Second, outside options generate negative spillovers that can trigger discontinuous exit cascades.¹⁷

¹⁷This mechanism is similar to the worsening adverse selection after a price increase in [Stiglitz and Weiss \(1981\)](#).

5.1.1 Comparative Statics on the Distribution of Types

We now consider how the distribution of types impacts the exit cascade result in Theorem 2. Specifically, how does the coarseness of agent types impact exit cascades? Let $\{I, m, u, c, \gamma\}$ represent an economy where $m = \{m_i\}_{i \in I}$, $u = \{u_i\}_{i \in I}$, $c = \{c_i\}_{i \in I}$, and $\gamma = \{\gamma_i\}_{i \in I}$. Consider another economy, $\{\hat{I}, \hat{m}, \hat{u}, \hat{c}, \hat{\gamma}\}$, that is a *coarser partition* of $\{I, m, u, c, \gamma\}$ if the following holds:

1. $\hat{I} \subset I$.
2. If $i \in \hat{I}$ and $i + 1 \in \hat{I}$, then $\hat{m}_i = m_i$, $\hat{u}_i = u_i$, and $\hat{c}_i = c_i$.
3. If $i \in \hat{I}$ and $i + 1, \dots, i + k \notin \hat{I}$, while $i + k + 1 \in \hat{I}$ or $i + k + 1 > n$, where $k \geq 1$, then agent $i \in \hat{I}$ includes agents $i, i + 1, \dots, i + k \in I$ and $\hat{m}_i = \sum_{l=0}^k m_{i+l}$, $\hat{u}_i(q, t) = \frac{\sum_{l=0}^k m_{i+l} u_{i+l}(q, t)}{\sum_{l=0}^k m_{i+l}}$, $\hat{c}_i = \frac{\sum_{l=0}^k m_{i+l} c_{i+l}}{\sum_{l=0}^k m_{i+l}}$, and $\hat{\gamma}_i = \frac{\sum_{l=0}^k m_{i+l} \gamma_{i+l}}{\sum_{l=0}^k m_{i+l}}$.

The above definition implies that a coarser partition of an economy groups adjacent types of agents into one type. The mass of the new type of agent is equal to the sum of all masses for each type in the group, and the servicing cost and outside option values are the weighted average cost and outside option values, respectively. The utility of this new agent is the weighted average utility across different types. Note that the upper-tail conditional expected cost is the same in coarse partition as the original fine partition.

Proposition 5 *Fix a type $i \in I$ whose outside option is strictly above autarky, $\gamma_i > u_i(0, 0)$, and suppose that **Condition ST**(i) holds at the pre-shock costs. Assume that type i 's participation threshold is reachable, i.e., there exists a price $p_i^+ > \bar{c}_i$ such that*

$$\max_{q \geq 0} u_i(q, p_i^+ q) \leq \gamma_i. \quad (18)$$

Let $\{\hat{I}, \hat{m}, \hat{u}, \hat{c}, \hat{\gamma}\}$ be a coarser partition of $\{I, m, u, c, \gamma\}$ with $i \in \hat{I}$ and $i + 1 \notin \hat{I}$, so that coarse type i contains at least types i and $i + 1$. Consider the participation

decision of type i conditional on all types below i being absent, so that type i is the lowest remaining type in both economies. Then, the following hold:

- (i) There exists an order-preserving cost vector $\tilde{c}$ with $\tilde{c}_j \geq c_j$ for all $j \in I$ such that type i exits in the fine economy $\{I, m, u, \tilde{c}, \gamma\}$, while coarse type i remains active in the coarse economy $\{\hat{I}, \hat{m}, \hat{u}, \hat{c}, \hat{\gamma}\}$.
- (ii) For any order-preserving cost vector $\tilde{c}$ with $\tilde{c}_j \geq c_j$ for all $j \in I$, if coarse type i exits in the coarse economy $\{\hat{I}, \hat{m}, \hat{u}, \hat{c}, \hat{\gamma}\}$, then type i exits in the fine economy $\{I, m, u, \tilde{c}, \gamma\}$.

This result shows that markets are less vulnerable to shutdowns and exit cascades when the partition of types becomes coarser. See Supplementary Appendix D for a visual illustration of the result.

We make two clarifications regarding the result and the proof. First, the proof evaluates only the mass-weighted *participation values* of the coarse type (the weighted average of its constituent types' utilities along the candidate layer and the weighted average of their outside options) and uses no further property of the averaged utility function.¹⁸ In particular, quasi-concavity or single crossing of $\hat{u}$ as a standalone utility function is never invoked. Second, the coarse economy is itself an instance of the model whenever the preference family is closed under mixing, as in the expected-utility insurance economy (Supplementary Appendix D provides an example). In the example, a coarse type's loss probability is the mass-weighted average of its members' probabilities, so the averaged preferences are again expected utility and inherit single crossing and quasi-concavity. For families that are not closed under mixing, the proposition should be read as a comparative static on the aggregated participation comparison of the constituent types.

¹⁸Moreover, the comparisons in the proposition are invariant to type-specific positive affine rescalings of utilities: both displays rest on the *within-type* signs of $u_l - \gamma_l$, term by term. At $\tilde{c}^*$, type i 's term is zero and, by condition (13), every higher term in the cell is strictly positive, so the mass-weighted comparison is positive under any rescaling. Conversely, a nonpositive weighted sum forces some nonpositive term, which by the contrapositive of (13) forces $u_i \leq \gamma_i$. The cardinal normalization of the averaged utility therefore plays no role in the results, only in the description of the coarse type as a standalone preference.

A corollary of Proposition 5 is a threshold comparative static: for the leading type (the lowest individual type for a given coarse grouping), the critical cost increase that triggers exit is weakly smaller under the finer partition than under the coarser one (coarse-partition exit implies fine-partition exit, not the reverse). Refining the type partition therefore weakly lowers the shock required to start a cascade, even with no change in underlying aggregate risk. This highlights why a multi-type model is essential for capturing large swings in trade from small cost increases. A model with only two types, while tractable and useful for illustrating certain forces, does not fully capture the market's vulnerability to exit cascades.

6 Market Shutdowns in Attar et al. (2014)

In this section we show that the model of Attar et al. (2014) does not generate market shutdowns after a small change in fundamentals, even when extended with generalized outside options.

A privately informed seller trades simultaneously with $n \geq 2$ identical uninformed buyers. No buyer can contract on the seller's trades with the others. The seller's type is L or H with probabilities m_L and m_H , and her preferences over aggregate bundles (Q, T) are represented by a continuously differentiable, strictly quasi-concave u_i with $\partial u_i / \partial T > 0$. Her marginal rate of substitution is $\tau_i \equiv -(\partial u_i / \partial Q) / (\partial u_i / \partial T)$, and strict single-crossing holds, $\tau_H(Q, T) > \tau_L(Q, T)$ for all (Q, T) , i.e., type H is less eager to sell an additional unit than type L . Buyers value the good at $v_L < v_H$, earn $v_i q - t$ on a trade (q, t) with type i , and the average quality is $v \equiv m_L v_L + m_H v_H \in (v_L, v_H)$. Each buyer posts a compact menu containing the null contract, and the seller then selects one contract from each menu. The equilibrium concept is pure-strategy perfect Bayesian equilibrium. Type i 's first-best quantity Q_i^* solves $\tau_i(Q_i^*, v_i Q_i^*) = v_i$.

Theorems 1 and 2 of Attar et al. (2014) establish that an equilibrium exists if and only if $\tau_L(0, 0) \leq v \leq \tau_H(0, 0)$, and characterize the equilibrium allocations by the position of v_L and v_H relative to the two types' marginal rates of substitution at $(0, 0)$:

- a. If $v_L \leq \tau_L(0, 0) \leq v \leq \tau_H(0, 0) \leq v_H$, all equilibria are pooling, with $Q_L = Q_H = 0$: at the pooling price v only type L would sell and only type H would buy, so no non-null contract survives.
- b. Otherwise all equilibria are separating, and the two types trade on opposite sides of the market at type-specific prices: (i) if $\tau_L(0, 0) < v_L < v < v_H < \tau_H(0, 0)$, then $Q_L = Q_L^* > 0$ and $Q_H = Q_H^* < 0$; (ii) if $\tau_L(0, 0) < v_L < v \leq \tau_H(0, 0) \leq v_H$, then $Q_L = Q_L^* > 0$ and $Q_H = 0$; (iii) if $v_L \leq \tau_L(0, 0) \leq v < v_H < \tau_H(0, 0)$, then $Q_L = 0$ and $Q_H = Q_H^* < 0$.

The feature that matters for our purposes is that no equilibrium exhibits partial pooling with both types trading on the same side: either both types are pooled at no trade, or each trades at a price determined by its own valuation. There is accordingly no shared layer whose price one type's exit could disturb.

In the following two subsections, we show that neither introducing small cost increases nor adding generalized outside options causes trades to completely unwind to zero. The main reason is that there is no partial pooling in the equilibrium contrary to the equilibrium in [Attar et al. \(2021\)](#). In equilibrium, either both types are already completely pooled into a no-trade contract or when one type is trading a non-zero quantity, the other type must trade in the opposite side of the market, i.e., one type is a buyer and the other type is a seller (including the possibility of trading zero quantities). Therefore, the terms of trade are not directly linked other than knife-edge cases.

6.1 Cost Increases Comparative Statics

Now we examine the effect of cost increases on the equilibrium quantities in the model of [Attar et al. \(2014\)](#). We first show that changes in the cost parameters of type L (of arbitrary size) do not affect the equilibrium quantity traded by type H , as long as an equilibrium continues to exist.

Proposition 6 (Robustness of the Attar et al. (2014) economy) *In the economy of Attar et al. (2014), consider an increase in m_L or a decrease in v_L of arbitrary size, and suppose $\tau_L(0,0) \neq v$ both before and after the change:*

- (i) *If the post-change parameters admit an equilibrium, type H 's equilibrium quantity of trade is unchanged, in particular, no such change drives H 's trade quantity to zero.*
- (ii) *If the post-change parameters do not admit an equilibrium, the failure occurs through nonexistence rather than through an exit cascade.*

Proposition 6 shows that changes in the cost of serving type L (of any size) leave the quantity of trade of type H unchanged as long as an equilibrium exists, showing a sharp contrast with the cascades of Section 5.1. Even though Proposition 6 applies to all cases in Theorem 1 and 2 in Attar et al. (2014), including the no-trade equilibrium, the rest of our analysis focuses on cases in which the equilibrium has non-zero trades.

6.2 Generalized Outside Options

We extend the model by adding outside options, which provides utility level of γ_L and γ_H for type L and H , respectively. If a seller chooses the outside option, then the seller cannot trade with any buyer. The extensive form of the game is the following:

1. After privately learning her type, the seller chooses whether to take the outside option or enter the market.
2. Each buyer k proposes a menu of contract C^k , which includes no trade, i.e., $(0,0)$.
3. If the seller chooses to enter the market, then the seller selects one contract from each of the menus C^k offered by the buyers.

We show that small changes in the cost parameters of type L do not affect the equilibrium trading decision of type H .

Proposition 7 *Suppose the generalized outside-option extension of the Attar et al. (2014) economy is in a separating equilibrium corresponding to one of cases b(i), b(ii), or b(iii). Then, for a sufficiently small decrease in v_L , no type H that participates before the change exits because of the change in type L 's terms or because type L exits. Hence, there is no exit cascades.*

The intuition for Proposition 7 explains why equilibrium structure is central to market fragility. In the separating equilibria of Attar et al. (2014), the two types never trade in the same direction—one type sells (positive quantity) while the other type buys (negative quantity), or one type exits. Thus, each type faces independent terms of trade, i.e., type L trades at unit price v_L , while type H trades at unit price v_H . When v_L decreases, this directly affects only the L type's trading decision and payoff, while the H type continues to face the same price v_H and the same contract terms, so H 's decision to trade or exit remains unchanged. Even if the L type exits the market due to the worse terms, L 's exit does not change the menu of contracts available to the H type unless it was prevented by L 's incentive compatibility condition.¹⁹

The results in Proposition 7 stand in sharp contrast to the exit cascade results established in Theorem 2 for the Attar et al. (2021) economy. In the extended model of Attar et al. (2021), equilibrium features partial pooling in which multiple types trade on the same side of the market and share common price layers in the convex market tariff.

When one type exits, the cheap shared layer supported by that type disappears from the choice set of the remaining types, lowering their market utility and potentially triggering further exits.

This direct linkage through shared prices is what enables small cost increases to trigger exit cascades. Thus, our results demonstrate that the presence of non-exclusive contracting and generalized outside options is insufficient to generate exit cascades and market shutdowns. Equilibrium requires pooling or cross-subsidization that creates strategic complementarities in exit decisions.

¹⁹Interestingly, L 's exit may open trade for type H . This is because outside options can serve as a secondary screening device.

The restriction to small changes in Proposition 7 preserves the initial equilibrium configuration, and it is not what shuts down the propagation channel. While our formal result is local, the reason the mechanism of Theorem 2 has no counterpart here is structural and does not depend on the shock size. Prices are type-specific rather than pooled: type H trades, when it trades, at v_H , which the shocks do not move, and type L 's presence affects H only through the incentive constraint that L 's contract imposes. A large decrease in v_L can push type L to its outside option directly, but this exit does not propagate, as it only relaxes that constraint, weakly opening trade for H (described in more detail in the next paragraph). Sequential, spillover-driven exits therefore cannot arise here, in contrast to what a pooled tariff transmits in Theorem 2.

Interestingly, introducing generalized outside options to the model of Attar et al. (2014) reveals a richer and perhaps surprising role beyond simply inducing exit cascades. They can act as a screening mechanism that enables trades previously unavailable.²⁰ For example, consider the pooling equilibrium case **a** in which neither type trades (both choose $(0, 0)$), because any attempt by buyers to offer contracts leads to adverse selection—both types want to mimic each other's contracts. However, when generalized outside options are introduced, if one type's outside option is sufficiently attractive relative to market trades (including mimicking a contract designed for the other type), it will voluntarily exit and eliminate the adverse selection problem. The remaining type can now trade at their type-specific price, as it is the only type active in the market. Thus, outside options can serve as an endogenous screening device similar to Board and Pycia (2014), as the willingness to forgo market participation reveals the type information, allowing buyers to safely offer contracts designed for the remaining type. Without the shared contract terms and potential cascade of exits shown in Theorem 2 under the extended model of Attar et al. (2021), the introduction of generalized outside options helps to maintain or revive trades, opening markets rather than shutting them down.

²⁰In Appendix C.3, we examine the case of no-trade (pooling) equilibrium in Attar et al. (2014) and show that generalizing outside options can result in an equilibrium with non-zero trades.

7 Discussion

We conclude with a discussion of empirical predictions our model generates. To connect them to observable markets, we interpret the outside option as certification: a participant can pay a fixed cost to have its quality verified and then trade on type-specific rather than pooled terms (see Appendix B). The mechanism therefore requires only a pooled market, an alternative that reveals quality, and a shock that changes the composition of the pool.

The same microfoundation delimits where the mechanism does *not* operate. Assumption 3 requires the value of certification, measured in units of the good, not to rise with type. It fails when the verified alternative is worth more to worse participants—as under actuarially fair insurance repricing, where the highest-risk types gain the most from coverage priced at their own rate. The predictions below therefore apply to markets in which separation benefits the strongest participants, such as structured products, credit ratings, and disclosure regimes, rather than to markets in which the weakest gain most from being priced individually.

Prediction 1 (Discontinuity). Theorem 2 predicts discontinuous responses to continuous deterioration in fundamentals. Observable market terms and movement toward the certified alternative should therefore adjust in steps rather than smoothly.

Agency mortgage-backed securities (MBS) provide a natural application. Most agency MBS trade in the to-be-announced (TBA) market, where collateral is specified only by broad characteristics and is therefore priced on pooled terms. Higher-quality collateral can instead be traded as a specified pool and priced on its own characteristics, at the cost of lower liquidity (Huh and Kim, 2021). The specified-pool pay-up relative to TBA prices therefore provides an observable counterpart to the value of leaving the pool. The model predicts that this premium, and the share of collateral moving outside the TBA pool, should respond discontinuously as pooled collateral quality deteriorates. A similar prediction applies to credit ratings: deterioration in an unrated pool should generate discrete increases in certification and in the pricing penalty borne by issuers that remain unrated (Peppe and Unal, 2025; Sufi, 2009).

Prediction 2 (Ordered exit). The cutoff property of Proposition 2 implies that the best, lowest-cost participants leave the pooled market first. Exit should therefore be ordered by observable quality rather than occurring uniformly across participants. In the mortgage application, higher-quality collateral should migrate to specified-pool trading before lower-quality collateral. This ordering also provides an early-warning implication, which is deterioration in the pool need not begin with the weakest participants leaving, but the departure of its strongest members can be the event that initiates the unraveling.

Prediction 3 (Amplification). The total contraction in a pooled market can exceed the direct effect of the shock that initiates it. The empirical challenge is therefore to distinguish participants who leave directly because of deteriorating fundamentals from those whose departure is induced by earlier exits. A credible test would compare the observed contraction with the direct response predicted from the initial shock alone, or exploit a shock that affects the participation margin of a particular group while leaving others' fundamentals unchanged.

The amplification result also has a policy implication. If an intervention prevents the triggering type from exiting, Corollary 2 implies that it prevents the subsequent cascade whenever the successor types remain strictly inside their spillover bands. A subsidy sufficient to preserve the first layer, of total cost $(\tilde{c}_1 - \bar{c}_1) q_1$ under the normalization $\sum_{i \in I} m_i = 1$, can therefore prevent a much larger contraction in trade. This is an amplification argument rather than a welfare theorem: because suppliers break even in any equilibrium, buyers' mass-weighted utilities capture the gains from trade, but without additional cardinal or quasi-linear structure the resource cost of the intervention cannot be compared directly with those utility gains.

Prediction 4 (Granularity). Proposition 5 implies that more precise classification can make pooled trade more fragile rather than more stable. Holding underlying risk fixed, markets that divide participants into more finely distinguished categories should therefore reach participation thresholds after smaller shocks. The mechanism is a cushioning effect: within a coarse category, the marginal participation surplus of

the strongest member is averaged with the strictly positive surpluses of the weaker members it is grouped with, so the category as a whole remains active after a shock that would have driven out its strongest member had that member been priced separately. Rating scales and disclosure tiers provide natural empirical counterparts, and the prediction supplies a static connection to the information-production literature (Gorton and Ordoñez, 2019, 2020; Dang et al., 2020).

Prediction 5 (Cheaper certification destabilizes). A lower cost of certification raises the value of the outside option and therefore makes exit easier. Holding pooled-market fundamentals fixed, improvements that make type-specific trading or verification cheaper should increase the fragility of the pooled market. In agency MBS, for example, lower transaction or liquidity costs in specified-pool trading should make high-quality collateral more willing to leave the TBA pool. The same logic applies more broadly to rating, disclosure, or third-party verification technologies: reducing the cost of distinguishing high-quality participants can worsen adverse selection among those who remain pooled.

These predictions suggest three especially direct empirical tests. First, one can examine whether departures from pooled markets are ordered by quality. Second, one can test for discrete changes in pooled-market pricing or certification around otherwise continuous deterioration in fundamentals. Third, variation in the cost or precision of certification can be used to test whether easier separation destabilizes the residual pooled market.

Appendix

A Omitted Proofs

A.1 Proof of Theorem 1

Proof. Statements (i) and (iv) are trivial as explained in the discussion before the theorem. At complete shutdown, the tariff is degenerate, so $\tau_i^{\tilde{T}}(0, 0) = \tau_i(0, 0)$, and

the condition is **Condition EP** for the post-shock inactive market (Theorem 1 of [Attar et al., 2021](#)).

For Statement (ii), first, consider type 1 and suppose, without loss of generality, that agent 1 was trading a positive quantity $q_1 > 0$. If type 1 exits the market after the cost increase, then $\tilde{q}_1 = 0 < q_1$ and the market tariff satisfies $\tilde{T}(q) \geq \tilde{c}_2 q \geq \bar{c}_2 q > \bar{c}_1 q$ for any q in $[0, q_1]$ and $[q_1, q_2]$ (that lies in the domain of $\tilde{T}$), regardless of the new trading quantities $\tilde{q}_2, \dots, \tilde{q}_n$. If instead type 1 still trades a positive quantity, the first layer's price rises to $\tilde{c}_1 \geq \bar{c}_1$, and given the new quantity $\tilde{q}_1$, type 2 faces

$$\frac{\tilde{T}(\tilde{q}_2) - \tilde{T}(\tilde{q}_1)}{\tilde{q}_2 - \tilde{q}_1} = \tilde{c}_2 \geq \bar{c}_2 = \frac{T(q_2) - T(q_1)}{q_2 - q_1},$$

whenever $\tilde{q}_2 > \tilde{q}_1$. Proceeding to an arbitrary type i , the same computation gives

$$\frac{\tilde{T}(\tilde{q}_i) - \tilde{T}(\tilde{q}_{i-1})}{\tilde{q}_i - \tilde{q}_{i-1}} = \tilde{c}_i \geq \bar{c}_i = \frac{T(q_i) - T(q_{i-1})}{q_i - q_{i-1}},$$

with the convention $\tilde{q}_0 \equiv 0$. Thus, every active layer of the new tariff is priced at the new, weakly higher upper-tail cost.

For Statement (iii), we first note that strict inequality in (11) holds if and only if the *layer gap*

$$G_i \equiv \max_{q' \geq 0} u_i(q_{i-1} + q', T(q_{i-1}) + \bar{c}_i q') - u_i(q_{i-1}, T(q_{i-1}))$$

is strictly positive. If $\tau_i(q_{i-1}, T(q_{i-1})) > \bar{c}_i$, some positive trade at a price strictly above $\bar{c}_i$ yields a strict gain, and the same trade at the lower price $\bar{c}_i$ yields a strictly larger one, so $G_i > 0$. Conversely, a strict gain at price $\bar{c}_i$ persists at price $\bar{c}_i + \eta$ for small $\eta > 0$ by continuity of u_i , so $\tau_i(q_{i-1}, T(q_{i-1})) \geq \bar{c}_i + \eta > \bar{c}_i$.

Next, we show that the post-shock equilibrium converges to the pre-shock equilibrium as the shock decreases to zero. Each layer's problem has a jointly continuous objective and a nonempty and bounded solution set, so its maximizers may be sought on a common compact interval for all cost vectors in a neighborhood of c . Then, by strict quasi-concavity, the solution is unique, and by Berge's maximum theorem, it is

continuous in the layer's starting point and price, with a continuous value. Consider layer 1: its starting point is $(0,0)$ and its price $\tilde{c}_1$ is continuous in $\tilde{c}$, so $\tilde{q}_1(\tilde{c}) \rightarrow q_1$ as $\tilde{c} \rightarrow c$. Consequently, layer 2's starting point $(\tilde{q}_1, \tilde{T}(\tilde{q}_1)) = (\tilde{q}_1, \tilde{c}_1 \tilde{q}_1)$ converges to $(q_1, T(q_1))$, and the argument propagates through the finitely many layers by induction. Each layer's starting point and price converge to their pre-shock values.

Finally, each layer gap is a continuous function of the layer's starting point and price, and equals $G_i > 0$ at the pre-shock equilibrium for every active layer by the assumption that the pre-shock market is active with slack. If no uniform $\bar{\epsilon}$ existed, there would be a sequence of shocks vanishing in $\max_i \epsilon_i$ and post-shock equilibria along which some fixed active layer's gap were nonpositive. However, along that sequence the layer's starting point and price converge to their pre-shock values, so its gap converges to $G_i > 0$, a contradiction. Hence, there exists $\bar{\epsilon} > 0$ such that all shocks with $\max_{i \in I} \epsilon_i < \bar{\epsilon}$ leave every pre-shock active layer with a strictly positive gap, that is, $\tilde{q}_i > \tilde{q}_{i-1}$, in every post-shock equilibrium. ■

A.2 Proof of Proposition 1

Proof. First, we prove that the active set is an upper interval. Suppose type i participates, trading a bundle $(\tilde{q}_i, \tilde{T}(\tilde{q}_i))$ with $u_i(\tilde{q}_i, \tilde{T}(\tilde{q}_i)) \geq u_i(0,0)$ and $\tilde{q}_i > 0$. By strict single-crossing, $u_j(\tilde{q}_i, \tilde{T}(\tilde{q}_i)) > u_j(0,0)$ for every $j > i$. Since the bundle lies on the tariff, it is feasible for j , so $\max_q u_j(q, \tilde{T}(q)) > u_j(0,0)$ and j participates. By contraposition, if a type does not trade, no lower type trades. Thus, the active set is an upper interval.

Next, we prove that the lowest types exit first. The marginal rate of substitution $\tau_i(q, t)$ is nonincreasing in q and u_i is continuous and decreasing in t . By continuity, there exists a cost vector $(\tilde{c}_j)_{j \in I}$ such that $u_{i-1}(0,0) > \max_q u_{i-1}(q, \tilde{T}(q))$, while $u_i(0,0) < \max_q u_i(q, \tilde{T}(q))$, i.e., type $i-1$ exits while type i remains. By the upper-interval property, every type $j < i-1$ then exits as well. ■

A.3 Proof of Proposition 2

Proof. Suppose type i participates, so that $\max_q u_i(q, T(q)) > \gamma_i \geq u_i(0,0)$ and

any optimal quantity is strictly positive. Let $(q, T(q))$ with $q > 0$ be such a bundle. Condition (13) implies $u_j(q, T(q)) > \gamma_j$ for every $j > i$. Because the same bundle is feasible for every higher type, $\max_{\tilde{q}} u_j(\tilde{q}, T(\tilde{q})) > \gamma_j$, so every $j > i$ participates. Therefore, the active set is an upper interval. ■

A.4 Proof of Proposition 3

Proof.

(i) The entrants $I_A = \{\theta+1, \dots, n\}$ form a finite type space with costs $c_{\theta+1} < \dots < c_n$ on which single-crossing and Assumption 2 follow from the full economy. Because I_A is an upper interval, for every $i \in I_A$, the upper-tail conditional cost computed within I_A equals that in the full economy, $\mathbb{E}[c_j \mid j \geq i, j \in I_A] = \mathbb{E}[c_j \mid j \geq i] = \bar{c}_i$. Applying Attar et al. (2021)'s Theorem 2 (see Section 3) to the subeconomy I_A delivers existence and uniqueness of the budget-feasible, entry-proof tariff, which is exactly $T^{\theta+1}$. Therefore, the baseline characterization carries over once the participant set is fixed.

(ii) Clause (i) holds for participants by hypothesis and, for an exited type $j \leq \theta$, because Lemma 1(ii) gives $U_j^{\theta+1} \leq U_j^j \leq \gamma_j$: a type that prefers its outside option to the market in which it is the best remaining participant also prefers it to the market that survives its exit. Clause (ii) holds by part (i) shown in the first part of this proof. Clause (iii) is the hypothesis $U_i^i \leq \gamma_i$ for $i \leq \theta$. The converse is immediate from clause (iii) of Definition 1. ■

A.5 Proof of Proposition 4

Proof. By Proposition 2, any such active set is an upper interval $\{k, \dots, n\}$, with $k = n+1$ denoting the empty set. Suppose $k < k^*$. Then, k participates while $U_k^k \leq \gamma_k$ by the definition of k^* . Since the tariff for the active set $\{k, \dots, n\}$ is T^k , clause (i) and the tie-breaking rule give $U_k^k > \gamma_k$, a contradiction. Hence $k \geq k^*$, and $A^* = \{k^*, \dots, n\}$ is the largest such set.²¹ ■

²¹ A^* itself satisfies clauses (i) through (iii) by Corollary 1.

A.6 Proof of Lemma 1

Proof. The tariff T^k is convex with $T^k(0) = 0$ and with minimal slope $\tilde{c}_k$. Thus, $T^k(q) \geq \tilde{c}_k q$ for every q in its domain, with equality on the first layer $[0, q_k^k]$. Since u_j is strictly decreasing in the transfer, $u_j(q, T^k(q)) \leq u_j(q, \tilde{c}_k q)$ for every feasible q . Taking the maxima yields the first inequality in (ii). Next, because $\tilde{c}_1 < \dots < \tilde{c}_n$, each upper-tail conditional expected cost $\tilde{c}_s$ is strictly increasing in s . Hence, $\tilde{c}_j \leq \tilde{c}_k$ for $j \leq k$, and monotonicity of u_j in the transfer gives $\max_q u_j(q, \tilde{c}_k q) \leq \max_q u_j(q, \tilde{c}_j q)$.

For statement (i), condition 2 of the layered characterization of the full economy (Section 3) can be applied to the subeconomy $\{k, \dots, n\}$ with the starting point $(q_{k-1}^k, T^k(q_{k-1}^k)) = (0, 0)$. Therefore, $q_k^k \in \arg \max_{q \geq 0} u_k(q, \tilde{c}_k q)$, and $T^k(q_k^k) = \tilde{c}_k q_k^k$, as q_k^k is the right endpoint of the first layer. Thus, $U_k^k \geq u_k(q_k^k, T^k(q_k^k)) = \max_q u_k(q, \tilde{c}_k q)$, while the reverse inequality is the first bound in (ii) with $j = k$. The final claim follows from statements (i) and (ii) applied to j : $U_j^k \leq U_j^j \leq \gamma_j$. ■

A.7 Proof of Lemma 2

Proof. Denote $q^\dagger \equiv q_{j-1}^{j-1}$. Under T^{j-1} , type j 's layer problem is $\max_{x \geq 0} u_j(q^\dagger + x, \tilde{c}_{j-1} q^\dagger + \tilde{c}_j x)$. By condition 2 of the characterization of the layered tariffs, its maximum is attained at $x = q_j^{j-1} - q^\dagger$, a point on the tariff, so U_j^{j-1} is at least this maximum. If some maximizer q_j^j of $u_j(q, \tilde{c}_j q)$ satisfies $q_j^j \geq q^\dagger$, evaluate the layer problem at $x = q_j^j - q^\dagger$, i.e.,

$$U_j^{j-1} \geq u_j(q_j^j, \tilde{c}_j q_j^j - (\tilde{c}_j - \tilde{c}_{j-1}) q^\dagger) \geq u_j(q_j^j, \tilde{c}_j q_j^j) = U_j^j,$$

where the second inequality is strict if $q^\dagger > 0$, since u_j is strictly decreasing in the transfer and $\tilde{c}_{j-1} < \tilde{c}_j$. If instead every maximizer satisfies $q_j^j < q^\dagger$, the bundle $(q_j^j, \tilde{c}_{j-1} q_j^j)$ lies on the first layer of T^{j-1} , so $U_j^{j-1} \geq u_j(q_j^j, \tilde{c}_{j-1} q_j^j) \geq u_j(q_j^j, \tilde{c}_j q_j^j) = U_j^j$, strictly if $q_j^j > 0$. Finally, if $q_j^j = 0$, then $U_j^j = u_j(0, 0) \leq U_j^{j-1}$, since no trade is feasible under T^{j-1} . ■

A.8 Proof of Theorem 2

Proof.

Sufficiency: We start by showing the *sufficiency* part. Assume that **Conditions IT**, **MS**(i), and **ST**(i) hold. We show that $(\{i, \dots, n\}, T^i)$ is an equilibrium and that it exhibits an exit cascade up to type i .

Step 1: The active types and resulting market tariff, $(\{i, \dots, n\}, T^i)$, form an equilibrium. We verify the three clauses of Definition 1 to show that the given market tariff and entrants form an equilibrium. (If $i = n + 1$, the active set is empty, clauses (i)–(ii) are vacuous for the active side, and clause (iii) is verified by the argument below applied to every type.)

Clause (ii). By Proposition 3(i) applied to the upper interval $\{i, \dots, n\}$, the market tariff T^i is the unique budget-feasible, entry-proof tariff among the entrants.

Clause (i), active types. Let $q_i^* \equiv q_i^i$, i.e., the right endpoint of the first layer of T^i . By the layered characterization, $q_i^* \in \arg \max_{q \geq 0} u_i(q, \tilde{c}_i q)$ and $T^i(q_i^*) = \tilde{c}_i q_i^*$, so $u_i(q_i^*, \tilde{c}_i q_i^*) = U_i^i$ by Lemma 1(i). Since $U_i^i > \gamma_i \geq u_i(0, 0)$ by **Condition ST**(i), we have $q_i^* > 0$. Condition (13) then yields $u_j(q_i^*, \tilde{c}_i q_i^*) > \gamma_j$ for every $j > i$. Because the bundle $(q_i^*, \tilde{c}_i q_i^*)$ lies on the tariff, it is feasible for every participant, so $U_j^i \geq u_j(q_i^*, T^i(q_i^*)) > \gamma_j$ for every $j > i$, and $U_i^i > \gamma_i$ directly. Therefore, every type $j \geq i$ strictly prefers participation.²²

Clause (i), exited types. By Lemma 1(ii), $U_j^i \leq U_j^j \leq \gamma_j$ for every $j < i$, where the second inequality is **Condition IT** for $j = 1$ and the left inequality of **Condition MS**(i) for $1 < j < i$. Given the tie-breaking rule, every exited type therefore optimally remains out when facing T^i .

Clause (iii). Clause (iii) of Definition 1 requires $U_j^j \leq \gamma_j$ for every exited type $j \leq i - 1$, which is exactly **Condition IT** together with the left inequalities in **Condition MS**(i).

Step 2: the equilibrium exhibits an exit cascade up to type i . Clause (a) of Definition 3 holds by construction. For clause (b), fix $j \in \{2, \dots, i - 1\}$: the

²²This paragraph supplies the participation requirement of Definition 1(i) for the active types, which Proposition 3(ii) takes as given.

right inequality in **Condition MS**(i), $\gamma_j < U_j^{j-1}$, states that type j strictly prefers trading along T^{j-1} to its outside option, so each exit after the first is triggered by its predecessor's.

The equilibrium is, moreover, reached constructively: by Lemma 1(i), the exit test of the sequential exit process (Definition 2) at step $j \leq i - 1$ is precisely $U_j^j \leq \gamma_j$, so the process eliminates types $1, \dots, i - 1$ in order and, by **Condition ST**(i), stops at type i : $A^* = \{i, \dots, n\}$.

Necessity: Next we show the *necessity* part. Suppose an equilibrium (I_A, T) exhibits an exit cascade up to type i , so that $I_A = \{i, \dots, n\}$ and, by clause (ii) of Definition 1 and Proposition 3(i), $T = T^i$.

Condition ST(i). If $i \leq n$, type i participates, so Definition 1(i) and the tie-breaking rule give $U_i^i > \gamma_i$. For $i = n + 1$, the condition is vacuous.

Condition IT and the left inequalities of **MS**(i). Let $j < i$ be an exited type. Clause (iii) of Definition 1 gives $U_j^j \leq \gamma_j$ directly. Applying this at $j = 1$ yields **Condition IT**, and at $1 < j \leq i - 1$ the left inequalities of **Condition MS**(i).

Right inequalities of MS(i). By clause (b) of Definition 3, each $j \in \{2, \dots, i - 1\}$ strictly prefers trading along T^{j-1} to its outside option, that is, $\gamma_j < U_j^{j-1}$. ■

A.9 Proof of Corollary 1

Proof. Existence. The process terminates in at most n steps at $A^* = \{k^*, \dots, n\}$ (possibly empty, written $k^* = n + 1$), with $U_j^j \leq \gamma_j$ for all $j < k^*$ and, if $A^* \neq \emptyset$, $U_{k^*}^{k^*} > \gamma_{k^*}$. Step 1 of the sufficiency proof of Theorem 2, applied with $i = k^*$ (including $k^* = 1$, when no type exits), verifies Definition 1 for (A^*, T^{k^*}) . When $A^* = \emptyset$, no type participates and clause (iii) holds, because $U_j^j \leq \gamma_j$ for every j by construction of the process, and clause (i) holds because a deviating buyer finds no contracts posted and obtains $u_i(0, 0) \leq \gamma_i$.

Uniqueness. By Proposition 2, any equilibrium active set is an upper interval $\{k, \dots, n\}$ or empty (written $k = n + 1$). If nonempty, Definition 1(i) and the tie-breaking rule give $U_k^k > \gamma_k$, while clause (iii) gives $U_j^j \leq \gamma_j$ for every $j < k$. The empty set requires $U_j^j \leq \gamma_j$ for every j . If two distinct candidates $\{k_1, \dots, n\}$ and

$\{k_2, \dots, n\}$ with $k_1 < k_2 \leq n + 1$ were both equilibria, the second would require $U_{k_1}^{k_1} \leq \gamma_{k_1}$ and the first $U_{k_1}^{k_1} > \gamma_{k_1}$, a contradiction. Given the active set, the tariff is unique by Proposition 3(i). ■

A.10 Proof of Corollary 2

Proof. The map $\tilde{c} \mapsto U_j^j(\tilde{c}) = \max_q u_j(q, \tilde{c}_j q)$ is continuous, since $\tilde{c}_j$ is continuous in $\tilde{c}$ and the value is continuous by Berge's maximum theorem. The map $\tilde{c} \mapsto U_j^{j-1}(\tilde{c})$ is continuous because the first-layer solution $q_{j-1}^{j-1}(\tilde{c})$ is unique by strict quasi-concavity and hence continuous, and the layer-problem value is continuous in its starting point and price. By hypothesis, type 1 participates before the shock, $\gamma_1 < U_1^1(c)$, so a shock with $U_1^1(\tilde{c}) \leq \gamma_1$ is a genuine exit event. The remaining conditions hold at $\tilde{c}$ by the choice of $\bar{\epsilon}$. Therefore, the finitely many strict inequalities in the hypothesis define an open neighborhood of c , yielding $\bar{\epsilon} > 0$ that preserves them all. For any shock in that neighborhood with $U_1^1(\tilde{c}) \leq \gamma_1$, Conditions **IT**, **MS**(i), and **ST**(i) hold at $\tilde{c}$, and Theorem 2 delivers the cascade. ■

A.11 Proof of Proposition 5

Proof. Suppose coarse type i contains the fine types $i, i + 1, \dots, i + k$, where $k \geq 1$. Because aggregation does not change the total mass or total cost of the upper tail beginning at i , the upper-tail conditional expected cost faced by the leading type is the same as $\tilde{c}_i$ in the fine and coarse economies.

We first prove statement (ii). Suppose coarse type i exits. Then

$$\max_{q \geq 0} \left[\frac{\sum_{l=0}^k m_{i+l} u_{i+l}(q, \tilde{c}_i q)}{\sum_{l=0}^k m_{i+l}} \right] \leq \hat{\gamma}_i = \frac{\sum_{l=0}^k m_{i+l} \gamma_{i+l}}{\sum_{l=0}^k m_{i+l}}. \quad (19)$$

Suppose, toward a contradiction, that fine type i does not exit. Then

$$\max_{q \geq 0} u_i(q, \tilde{c}_i q) = u_i(q_i, \tilde{c}_i q_i) > \gamma_i$$

for some optimizer q_i . Since $\gamma_i > u_i(0, 0)$, $q_i > 0$. Condition (13) therefore implies $u_{i+l}(q_i, \tilde{c}_i q_i) > \gamma_{i+l}$, for $l = 1, \dots, k$. Together with $u_i(q_i, \tilde{c}_i q_i) > \gamma_i$, this gives

$$\begin{aligned} \max_{q \geq 0} \left[\frac{\sum_{l=0}^k m_{i+l} u_{i+l}(q, \tilde{c}_i q)}{\sum_{l=0}^k m_{i+l}} \right] &\geq \frac{\sum_{l=0}^k m_{i+l} u_{i+l}(q_i, \tilde{c}_i q_i)}{\sum_{l=0}^k m_{i+l}} \\ &> \frac{\sum_{l=0}^k m_{i+l} \gamma_{i+l}}{\sum_{l=0}^k m_{i+l}} = \hat{\gamma}_i, \end{aligned}$$

contradicting (19). Hence, coarse-type exit implies fine-type exit.

We next prove statement (i). Define the maximized utility given unit price p as

$$W_i(p) \equiv \max_{q \geq 0} u_i(q, pq).$$

By the maintained locally uniform demand-boundedness condition and Berge's Maximum Theorem, $W_i(p)$ is continuous in p on the compact interval $[\bar{c}_i, p_i^+]$. Since u_i is strictly decreasing in the transfer, $W_i(p)$ is weakly decreasing in p .

By **Condition ST**(i), $W_i(\bar{c}_i) > \gamma_i$, and by (18), $W_i(p_i^+) \leq \gamma_i$. Hence, by continuity, there exists $p_i^* \in (\bar{c}_i, p_i^+]$ such that

$$W_i(p_i^*) = \gamma_i. \tag{20}$$

Let $\Delta \equiv p_i^* - \bar{c}_i > 0$, and construct the cost vector²³

$$\tilde{c}_j^* = \begin{cases} c_j, & j < i, \\ c_j + \Delta, & j \geq i. \end{cases}$$

Note that $\tilde{c}_i^* = \bar{c}_i + \Delta = p_i^*$. Let q_i^* be an optimizer in (20). Since $u_i(q_i^*, p_i^* q_i^*) = \gamma_i > u_i(0, 0)$, we have $q_i^* > 0$. By the tie-breaking rule, fine type i exits at $\tilde{c}^*$.

On the other hand, condition (13) applied to the positive-quantity bundle $(q_i^*, p_i^* q_i^*)$

²³This cost change preserves the strict ordering of costs and satisfies $\tilde{c}_j^* \geq c_j$ for every j .

gives $u_{i+l}(q_i^*, p_i^* q_i^*) > \gamma_{i+l}$, for $l = 1, \dots, k$. Therefore,

$$\begin{aligned} \max_{q \geq 0} \left[\frac{\sum_{l=0}^k m_{i+l} u_{i+l}(q, p_i^* q)}{\sum_{l=0}^k m_{i+l}} \right] &\geq \frac{\sum_{l=0}^k m_{i+l} u_{i+l}(q_i^*, p_i^* q_i^*)}{\sum_{l=0}^k m_{i+l}} \\ &> \frac{\sum_{l=0}^k m_{i+l} \gamma_{i+l}}{\sum_{l=0}^k m_{i+l}} = \hat{\gamma}_i. \end{aligned}$$

Thus coarse type i strictly prefers participation at the same cost vector $\tilde{c}^*$, while fine type i exits. This proves statement (i). ■

A.12 Proof of Proposition 6

Proof. By Theorem 1 in [Attar et al. \(2014\)](#), equilibrium exists if and only if $\tau_L(0, 0) < v \leq \tau_H(0, 0)$ where we exclude the knife-edge case of $\tau_L(0, 0) = v$. Within this region, the four configurations—a, b(i)–(iii)—partition the parameter space according to the orderings of v_L versus $\tau_L(0, 0)$ and of v_H versus $\tau_H(0, 0)$, and type H 's trade is determined by the second ordering alone:

1. If $\tau_H(0, 0) \leq v_H$ (cases a and b(ii)), then $Q_H = 0$.
2. If $v_H < \tau_H(0, 0)$ (cases b(i) and b(iii)), then $Q_H = Q_H^* < 0$, where $\tau_H(Q_H^*, Q_H^* v_H) = v_H$.

Both the ordering and Q_H^* depend only on v_H and type H 's preferences. An increase in m_L or a decrease in v_L leaves v_H unchanged and weakly lowers $v = m_L v_L + m_H v_H$, so $v \leq \tau_H(0, 0)$ is preserved. Hence, as long as $\tau_L(0, 0) < \tilde{v}$ after the change, the configuration may transition (the ordering of $\tau_L(0, 0)$ and v_L may reverse, moving b(i) to b(iii) or b(ii) to a) but Q_H is unchanged.

If instead the change is large enough that $\tilde{v} < \tau_L(0, 0)$, no equilibrium exists: the failure occurs through the nonexistence channel of [Attar et al. \(2014\)](#) rather than through an exit of type H . ■

A.13 Proof of Proposition 7

Proof of Proposition 7.

We analyze the effect of a decrease in v_L to $\tilde{v}_L$ on equilibrium in the separating cases of Theorem 1 of Attar et al. (2014). The key observation is that in separating equilibria, the two types face *independent terms of trade* determined by their respective values v_L and v_H .

Step 1. All cases of equilibria.

Case b(i): $\tau_L(0, 0) < v_L < v < v_H < \tau_H(0, 0)$, with $Q_L = Q_L^* > 0$ (L sells) and $Q_H = Q_H^* < 0$ (H buys).

In equilibrium, buyers offer a contract for L , $(Q_L^*, v_L Q_L^*)$, where $\tau_L(Q_L^*, v_L Q_L^*) = v_L$, and a contract for H , $(Q_H^*, v_H Q_H^*)$, where $\tau_H(Q_H^*, v_H Q_H^*) = v_H$ and $Q_H^* < 0$.

When v_L decreases to $\tilde{v}_L$, the new equilibrium contract for L becomes $(\tilde{Q}_L^*, \tilde{v}_L \tilde{Q}_L^*)$ where $\tau_L(\tilde{Q}_L^*, \tilde{v}_L \tilde{Q}_L^*) = \tilde{v}_L$. The contract for H remains $(Q_H^*, v_H Q_H^*)$ since v_H is unchanged.

First, we check the incentive compatibility (IC) of the two types:

- H 's IC: $u_H(Q_H^*, v_H Q_H^*) \geq u_H(\tilde{Q}_L^*, \tilde{v}_L \tilde{Q}_L^*)$

Since the initial separating equilibrium satisfies H 's incentive constraint with strict slack, and both sides of the constraint are continuous in v_L , a sufficiently small decrease in v_L preserves it.

Thus, by continuity, H 's incentive-compatibility constraint remains satisfied for all sufficiently small decreases in v_L .

- L 's IC: $u_L(\tilde{Q}_L^*, \tilde{v}_L \tilde{Q}_L^*) \geq u_L(Q_H^*, v_H Q_H^*)$

For sufficiently small decreases in v_L , this constraint remains satisfied since initially there was slack, $u_L(Q_L^*, v_L Q_L^*) > u_L(Q_H^*, v_H Q_H^*)$, in equilibrium.

Next, we check the effect of outside options:

- L exits if $u_L(\tilde{Q}_L^*, \tilde{v}_L \tilde{Q}_L^*) < \gamma_L$, which is possible.
- H exits if $u_H(Q_H^*, v_H Q_H^*) < \gamma_H$, which is unchanged from before.

Therefore, for sufficiently small decreases in v_L , H 's trading decision is completely unaffected. L may exit, but this does not cause H to exit. Thus, there are no cascades.

Case b(ii): $\tau_L(0,0) < v_L < v \leq \tau_H(0,0) \leq v_H$, with $Q_L = Q_L^* > 0$ and $Q_H = 0$.

H is not trading because either (i) $\gamma_H > u_H(Q_H^*, v_H Q_H^*)$ for any positive trading quantity, or (ii) the market structure does not support H 's trade.

When v_L decreases to $\tilde{v}_L$ L 's contract changes to $(\tilde{Q}_L^*, \tilde{v}_L \tilde{Q}_L^*)$. In contrast, the value v_H and H 's potential trading terms are unchanged. Therefore H 's decision to not trade remains unchanged if L remains to trade.

If L exits (when $u_L(\tilde{Q}_L^*, \tilde{v}_L \tilde{Q}_L^*) < \gamma_L$), the equilibrium depends on the H 's incentive to enter the market and L 's incentive to mimic H .

- If $\gamma_L < u_L(Q_H^*, v_H Q_H^*)$, where $Q_H^* > 0$ because of $\tau_H(0,0) < v_H$, then both types exit the market in equilibrium. In other words, because of L type's incentive to mimic H , a contract designed for H would not be traded in equilibrium.
- If $\gamma_L > u_L(Q_H^*, v_H Q_H^*)$ and $\gamma_H < u_H(Q_H^*, v_H Q_H^*)$, then L exits while H enters the market. In other words, if L exits, then a contract designed for H could be traded, as there is no adverse selection problem.

Therefore, a decrease in v_L does not trigger an additional exit of H , i.e., there are no cascades.

Case b(iii): $v_L \leq \tau_L(0,0) \leq v < v_H < \tau_H(0,0)$, with $Q_L = 0$ and $Q_H = Q_H^* < 0$.

L is not trading because $v_L \leq \tau_L(0,0)$, i.e., selling is not profitable.

When v_L decreases to $\tilde{v}_L < v_L$, the condition $\tilde{v}_L < v_L \leq \tau_L(0,0)$ still holds, thus, L remains unwilling to trade (even more so now). H 's contract $(Q_H^*, v_H Q_H^*)$ is determined by v_H , which is unchanged, therefore $u_H(Q_H^*, v_H Q_H^*)$ is unchanged, so H 's decision is unchanged.

Step 2. v_L comparative statics.

Now, we show that under any possible setup for outside options, H 's incentives and equilibrium contracts are unaffected by the decrease in v_L , and, therefore, there are no cascades.

We first introduce generalized outside options to the cases b. (i) through b. (iii) in [Attar et al. \(2014\)](#) and show the impact on equilibrium trades. In general, there are two trivial extreme cases: (1) both outside options are too low, so they are irrelevant; (2) both outside options are too high, so mechanically induce exits. In the intermediate cases, outside options can act as a screening mechanism. For example, it is possible that H type does not trade in the original equilibrium in [Attar et al. \(2014\)](#), however, the L type exits after an introduction of outside options, eliminating the asymmetric information problem and incentivizing the H type to enter and trade. The following is the exhaustive list of all cases and parameter configurations:

1. If $\tau_L(0, 0) < v_L < v < v_H < \tau_H(0, 0)$, hence $Q_L = Q_L^* > 0$, $Q_H = Q_H^* < 0$.
 - (a) If $\gamma_L < u_L(Q_L^*, Q_L^* v_L)$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium trivially remains the same as in that in Theorem 1 of [Attar et al. \(2014\)](#).
 - (b) If $\gamma_L > u_L(Q_L^*, Q_L^* v_L)$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is separating with L type exiting and H type trading $Q_H^* < 0$.
 - (c) If $\gamma_L < u_L(Q_L^*, Q_L^* v_L)$ and $\gamma_H > u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is separating with H type exiting and L type trading $Q_L^* > 0$.
 - (d) If $\gamma_L > u_L(Q_L^*, Q_L^* v_L)$ and $\gamma_H > u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is the pooling equilibrium with both types exiting the market.
2. If $\tau_L(0, 0) < v_L < v \leq \tau_H(0, 0) \leq v_H$, hence $Q_L = Q_L^* > 0$, $Q_H = 0$.
 - (a) If $\gamma_L < u_L(Q_L^*, Q_L^* v_L)$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, where $Q_H^* > 0$ such that $\tau_H(Q_H^*, Q_H^* v_H) = v_H$. Then, the equilibrium trivially remains the same as in that in Theorem 1 of [Attar et al. \(2014\)](#).
 - (b) If $u_L(Q_L^*, Q_L^* v_L) < \gamma_L < u_L(Q_H^*, Q_H^* v_H)$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is the pooling equilibrium with both types exiting the market.
 - (c) If $u_L(Q_L^*, Q_L^* v_L) < u_L(Q_H^*, Q_H^* v_H) < \gamma_L$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is a separating equilibrium with L type exiting and H type trading $Q_H^* > 0$.

- (d) If $\gamma_L < u_L(Q_L^*, Q_L^* v_L)$ and $\gamma_H > u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is separating with H type exiting and L type trading $Q_L^* > 0$.
 - (e) If $\gamma_L > u_L(Q_L^*, Q_L^* v_L)$ and $\gamma_H > u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is the pooling equilibrium with both types exiting the market.
3. If $v_L \leq \tau_L(0, 0) \leq v < v_H < \tau_H(0, 0)$, hence $Q_L = 0$ and $Q_H = Q_H^* < 0$.
- (a) If $\gamma_L < u_L(Q_L^*, Q_L^* v_L)$, where $Q_L^* < 0$ such that $\tau_L(Q_L^*, Q_L^* v_L) = v_L$, and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, where $Q_H^* < 0$ such that $\tau_H(Q_H^*, Q_H^* v_H) = v_H$. Then, the equilibrium trivially remains the same as in that in Theorem 1 of [Attar et al. \(2014\)](#).
 - (b) If $\gamma_L < u_L(Q_L^*, Q_L^* v_L)$ and $u_H(Q_H^*, Q_H^* v_H) < \gamma_H < u_H(Q_L^*, Q_L^* v_L)$, then the equilibrium is the pooling equilibrium with both types exiting the market.
 - (c) If $\gamma_L < u_L(Q_L^*, Q_L^* v_L)$ and $u_H(Q_H^*, Q_H^* v_H) < u_H(Q_L^*, Q_L^* v_L) < \gamma_H$, then the equilibrium is a separating equilibrium with L type trading $Q_L^* < 0$ and H type exiting the market.
 - (d) If $\gamma_L > u_L(Q_L^*, Q_L^* v_L)$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is separating with L type exiting and H type trading $Q_H^* < 0$.
 - (e) If $\gamma_L > u_L(Q_L^*, Q_L^* v_L)$ and $\gamma_H > u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is the pooling equilibrium with both types exiting the market.

Now, we change v_L to $\tilde{v}_L$ and examine what happens to the equilibrium in every case listed above.

- 1. If $\tau_L(0, 0) < v_L < v < v_H < \tau_H(0, 0)$, hence $Q_L = Q_L^* > 0$, $Q_H = Q_H^* < 0$. In any of the cases, a decrease in v_L to $\tilde{v}_L$ may induce an exit of L type, but the H 's decision remains the same.
 - (a) If $\gamma_L < u_L(Q_L^*, Q_L^* \tilde{v}_L)$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium trivially remains the same as in that in Theorem 1 of [Attar et al. \(2014\)](#).
 - (b) If $\gamma_L > u_L(Q_L^*, Q_L^* \tilde{v}_L)$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is separating with L type exiting and H type trading $Q_H^* < 0$.

- (c) If $\gamma_L < u_L(Q_L^*, Q_L^* \tilde{v}_L)$ and $\gamma_H > u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is separating with H type exiting and L type trading $Q_L^* > 0$.
- (d) If $\gamma_L > u_L(Q_L^*, Q_L^* \tilde{v}_L)$ and $\gamma_H > u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is the pooling equilibrium with both types exiting the market.
2. If $\tau_L(0, 0) < v_L < v \leq \tau_H(0, 0) \leq v_H$, hence $Q_L = Q_L^* > 0$, $Q_H = 0$. In any of the cases, a decrease in v_L to $\tilde{v}_L$ may induce an exit of L type, but the H 's decision remains the same.
- (a) If $\gamma_L < u_L(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L)$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, where $Q_H^* > 0$ such that $\tau_H(Q_H^*, Q_H^* v_H) = v_H$. Then, the equilibrium trivially remains the same as in that in Theorem 1 of [Attar et al. \(2014\)](#).
- (b) If $u_L(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L) < \gamma_L < u_L(Q_H^*, Q_H^* v_H)$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is the pooling equilibrium with both types exiting the market.
- (c) If $u_L(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L) < u_L(Q_H^*, Q_H^* v_H) < \gamma_L$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is a separating equilibrium with L type exiting and H type trading $Q_H^* > 0$.
- (d) If $\gamma_L < u_L(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L)$ and $\gamma_H > u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is separating with H type exiting and L type trading $Q_L^* > 0$.
- (e) If $\gamma_L > u_L(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L)$ and $\gamma_H > u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is the pooling equilibrium with both types exiting the market.
3. If $v_L \leq \tau_L(0, 0) \leq v < v_H < \tau_H(0, 0)$, hence $Q_L = 0$ and $Q_H = Q_H^* < 0$. Again, in any of the cases, a decrease in v_L to $\tilde{v}_L$ does not induce an exit of H .
- (a) If $\gamma_L < u_L(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L)$, where $\tilde{Q}_L^* < 0$ such that $\tau_L(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L)$, and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, where $Q_H^* < 0$ such that $\tau_H(Q_H^*, Q_H^* v_H) = v_H$. Then, the equilibrium trivially remains the same as in that in Theorem 1 of [Attar et al. \(2014\)](#).

- (b) If $\gamma_L < u_L(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L)$ and $u_H(Q_H^*, Q_H^* v_H) < \gamma_H < u_H(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L)$, then the equilibrium is the pooling equilibrium with both types exiting the market.
- (c) If $\gamma_L < u_L(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L)$ and $u_H(Q_H^*, Q_H^* v_H) < u_H(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L) < \gamma_H$, then the equilibrium is a separating equilibrium with L type trading $\tilde{Q}_L^* < 0$ and H type exiting the market.
- (d) If $\gamma_L > u_L(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L)$ and $\gamma_H < u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is separating with L type exiting and H type trading $Q_H^* < 0$.
- (e) If $\gamma_L > u_L(\tilde{Q}_L^*, \tilde{Q}_L^* \tilde{v}_L)$ and $\gamma_H > u_H(Q_H^*, Q_H^* v_H)$, then the equilibrium is the pooling equilibrium with both types exiting the market.

Step 3. Conclusion

In all separating equilibria, changes in v_L do not cause H to exit through a cascade mechanism because:

1. H 's terms of trade are determined solely by v_H , which is unchanged.
2. In separating equilibria, types face independent pricing.
3. L 's exit (if it occurs) does not change H 's available contracts or payoffs.

This contrasts sharply with [Attar et al. \(2021\)](#) where partial pooling creates direct linkages: when one type exits a pool, the pooling price changes for all remaining types, creating cascade dynamics. ■

B Microfoundation of Outside Options

Though sufficient, condition (13) is not necessary for exits to generate additional exits due to negative spillovers. However, without some structure, the additional complexity of [Rothschild and Stiglitz \(1976\)](#)-type of non-existence problems arise. In addition, the pattern of exits could become more complicated as there might be no cutoff type, θ , that partitions agents into those who exit and remain. In particular, some intermediate valued agent may remain in the market if the agent's outside option utility is very low.

We now provide a microfoundation of outside options that justifies assumption (13). We interpret outside options as the opportunity cost of trading in a separate market that verifies each agent's type at a fixed cost. In this verified market, an agent pays a fee $\kappa > 0$ and then trades at a type-specific unit price equal to its own cost, so that

$$\gamma_i = \max_{q' \geq 0} u_i(q', c_i q' + \kappa), \quad (21)$$

with maximizer q_i^v . We consider fee levels at which verification is weakly worthwhile for every type, so that $\gamma_i \geq u_i(0, 0)$ as maintained throughout the paper.

To compare utility under verification across types without imposing cardinal structure, we measure them in units of the good. Since $u_i(q_i^v, 0) > u_i(q_i^v, c_i q_i^v + \kappa) = \gamma_i \geq u_i(0, 0)$ and u_i is continuous, there exists a smallest $\underline{q}_i \in [0, q_i^v)$ such that

$$u_i(\underline{q}_i, 0) = \gamma_i, \quad (22)$$

i.e., the free quantity that exactly compensates type i for forgoing verification. By minimality, $u_i(q, 0) \leq \gamma_i$ for every $q \leq \underline{q}_i$. Finally, we assume the following:

Assumption 3 *The compensating quantities are nonincreasing in type: $\underline{q}_j \leq \underline{q}_i$ for all $i < j$.*

Assumption 3 states that, measured in units of the good, the value of the verified option does not rise with type, i.e., a worse type's price disadvantage in the verified market ($c_j > c_i$) weakly outweighs its stronger desire to trade there. It is a genuine restriction, as the two forces push in opposite directions, and it delimits where the verification interpretation applies.

Lemma B.1 (Verification microfoundation) *Let γ_i be given by (21) and suppose Assumption 3 holds. Then, for all $i < j$, condition (13) holds at every bundle (q, t) with $q > \underline{q}_j$. In particular, it holds at every bundle with a strictly positive transfer.*

Proof. Fix $i < j$ and a bundle (q, t) with $q > \underline{q}_j$ and $u_i(q, t) \geq \gamma_i$. Since $\underline{q}_j \leq \underline{q}_i$, minimality of $\underline{q}_i$ gives $u_i(\underline{q}_i, 0) \leq \gamma_i \leq u_i(q, t)$. Strict single-crossing applied to the

pair $(\underline{q}_j, 0)$ and (q, t) , with $\underline{q}_j < q$, then yields $u_j(q, t) > u_j(\underline{q}_j, 0) = \gamma_j$, as required. For the final claim, if $t > 0$ and $q \leq \underline{q}_j \leq \underline{q}_i$, then $u_i(q, t) < u_i(q, 0) \leq \gamma_i$, so the premise of (13) cannot hold. The restriction $q > \underline{q}_j$ is therefore vacuous for positively priced bundles. ■

The main cascade argument invokes condition (13) to propagate participation across positive-trade market bundles. Along every nondegenerate entry-proof layer, quantities and transfers are strictly positive because unit costs are strictly positive. Therefore, Lemma B.1 provides a concrete verification-market microfoundation for the positive-transfer comparisons that drive the cascade. Under quasi-linear preferences, $u_i(q, t) = v_i(q) - t$ with $v_i(0) = 0$, Assumption 3 is implied by (and is strictly weaker than) monotonicity of the verified surplus, $\max_q \{v_j(q) - c_j q\} \leq \max_q \{v_i(q) - c_i q\}$ for $i < j$.

The verification fee is a comparative static of independent interest. Since u_i is strictly decreasing in the transfer, γ_i in (21) is strictly decreasing in κ for every i : a cheaper certification technology raises every type's outside option. Holding costs fixed, this relaxes **Condition IT** and the left inequalities of **Condition MS**(i), so both the initial exit and each subsequent one require a smaller cost increase. By Corollary 2, the shock sufficient to trigger the full cascade shrinks accordingly. Cheaper verification therefore destabilizes the pooled market rather than supporting it.²⁴

Finally, one can think of a general equilibrium model in which the utility levels of outside options are endogenously affected by entry/exit decisions of all agents.²⁵ Our exit cascade results continue to hold even if the value of outside options declines as the best types exit the market. More specifically, exit cascades occur as long as the downward jump in market utility for remaining agents is larger than the decline in value of the outside option. Such general equilibrium interactions are out of the scope

²⁴The comparison is made at fee levels for which Assumption 3 continues to hold, since a change in κ moves every compensating quantity $\underline{q}_i$.

²⁵The current formulation of outside options under the interpretation discussed above does not allow the exit decisions of agents to impact the value of others' outside options. For example, the fixed entry fee is independent of other agents' entry and exit decisions. Also, if the outside option "market" can identify the true type of each agent, entry of moldy lemons or other agents into this outside option market will not affect the utility of the lower type agents.

of this paper but will be an interesting direction for future research. Along similar lines, changes in outside option values themselves can generate exit cascades even without changes in costs. However, full-fledged comparative statics on the distribution of outside options would be highly complicated due to the strategic complementarity and substitutability issues discussed in the Supplementary Appendix [C.2](#).

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Supplementary Appendices

C Additional Results

C.1 Description of Cascade of Exits

We provide a detailed illustrative description of the exit cascade mechanism and general conditions on the size of outside options needed for exit cascades to occur. The inequalities below trace, layer by layer, the spillover established in Lemma 2, where each type's exit strictly worsens the market for its successor, and the size of the outside option determines whether that loss converts into a further exit. Assume a cost increase, $\tilde{c}_1$, that is sufficient to ensure condition (15) holds for agent 1 to exit the market. If

$$\max_{q \geq 0} u_2(q, \tilde{c}_2 q) \leq \gamma_2,$$

agent 2 will also exit. Importantly, agent 2's exit condition above can hold even if agent 2 enters the market conditional on agent 1 remaining:

$$\max_{q \geq 0} u_2(q_1 + q, \tilde{c}_1 q_1 + \tilde{c}_2 q) > \gamma_2,$$

where $q_1 = \arg \max_{q \geq 0} u_1(q, \tilde{c}_1 q)$, because

$$\max_{q \geq 0} u_2(q_1 + q, \tilde{c}_1 q_1 + \tilde{c}_2 q) > \max_{q \geq 0} u_2(q, \tilde{c}_2 q)$$

as long as $q_1 > 0$ (Lemma 2).

A similar but even more powerful mechanism exists for agent 3's entry/exit decision. The level of agent 3's outside option utility level may be low enough that she

will enter the market despite agent 1 exiting while agent 2 enters:

$$\max_{q \geq 0} u_3(q_2 + q, \tilde{c}_2 q_2 + \tilde{c}_3 q) > \gamma_3,$$

where $q_2 = \arg \max_{q \geq 0} u_2(q, \tilde{c}_2 q)$. This is because

$$\max_{q \geq 0} u_3(q_1 + (q_2 - q_1) + q, \tilde{c}_1 q_1 + \tilde{c}_2(q_2 - q_1) + \tilde{c}_3 q) > \max_{q \geq 0} u_3(q_2 + q, \tilde{c}_2 q_2 + \tilde{c}_3 q) > \max_{q \geq 0} u_3(q, \tilde{c}_3 q),$$

which implies that γ_3 can be between $\max_{q \geq 0} u_3(q_2 + q, \tilde{c}_2 q_2 + \tilde{c}_3 q)$ and $\max_{q \geq 0} u_3(q, \tilde{c}_3 q)$.

Extending this argument to any general i , inductively, we see that

$$\begin{aligned} & \max_{q \geq 0} u_i(q_1 + (q_2 - q_1) + \cdots + (q_{i-1} - q_{i-2}) + q, \tilde{c}_1 q_1 + \tilde{c}_2(q_2 - q_1) \cdots + \tilde{c}_{i-1}(q_{i-1} - q_{i-2}) + \tilde{c}_i q) \\ & > \max_{q \geq 0} u_i(q_2 + \cdots + (q_{i-1} - q_{i-2}) + q, \tilde{c}_2 q_2 + \cdots + \tilde{c}_{i-1}(q_{i-1} - q_{i-2}) + \tilde{c}_i q) \\ & \quad \vdots \\ & > \max_{q \geq 0} u_i(q, \tilde{c}_i q), \end{aligned}$$

so γ_i can be as low as an element of an ϵ -neighborhood of $\max_{q \geq 0} u_i(q, \tilde{c}_i q)$.

The key insight of the exit cascade argument is that the only outside option value that needs to be close to an agent's maximal attainable utility is that of the best type who exits first. Conditional on the first agent's exit, the downward jumps in utility accumulate, meaning that agent n faces all the utility decreases that arise from agents' $1, 2, \dots, n-1$ exiting. If we consider a model where the set of eligible participants is endogenous (e.g., agent 1 is the marginal type who is close to being indifferent between entering and exiting the market), the condition that agent 1's outside option utility level be sufficiently close to their ex ante equilibrium utility level arises naturally. Therefore, the required parametric conditions may not be as restrictive as they appear.

C.2 Rothschild-Stiglitz Unraveling: Nonexistence

This subsection revisits the existence arguments of [Attar et al. \(2021\)](#) in light of introducing outside options. Recall in their model, equilibrium always exists and is unique. We now show that our generalization of outside options also alters the existence result of the model, similar to the [Rothschild and Stiglitz \(1976\)](#) unraveling argument with exclusive contracting.

A necessary condition for the non-existence of equilibrium is that assumption (13) does not hold.²⁶ The following simple example highlights how outside options can cause equilibrium to break down when (13) does not hold. Suppose that type n agent exits the market because the aggregate market tariff T is too high compared to n 's outside option, i.e. $\max_q u_n(q, T(q)) \leq \gamma_n$. However, the exit of n decreases T to $T^\dagger$, which may make n want to enter again. Then, the entry of n will change (increase) $T^\dagger$ to T again, which will make n want to exit again.

Furthermore, an agent's entry or exit could affect other agents' entry and exit decisions, further complicating equilibrium existence. Suppose the cost vector increases to $\tilde{c} > c$. Without loss of generality, we assume that all agents enter the market in the equilibrium before the cost increase. As in the previous section, the new upper-tail conditional expected cost is given by (10). Denote equilibrium quantities by $\tilde{q}_i$ for each i that enters the market where

$$\tilde{q}_i - \tilde{q}_{i-1} \in \arg \max_q \left\{ u_i \left(\tilde{q}_{i-1} + q, \tilde{T}(\tilde{q}_{i-1}) + \tilde{c}_i q \right) : q \right\},$$

and $\tilde{T}(q)$ is the equilibrium aggregate affine tariff with slope $\tilde{c}_i$ over $[\tilde{q}_{i-1}, \tilde{q}_i]$. In addition, set $\tilde{q}_0 = 0$ and $\tilde{q}_j = \tilde{q}_{j-1}$ for any j who exits the market, so that an exiting type has an empty layer.

In this case, agent entry and exit decisions in equilibrium may be either strategic complements or substitutes à la [Bulow et al. \(1985\)](#). Suppose that type i agent does

²⁶Theorem 1 of [Attar et al. \(2014\)](#) shows that an equilibrium exists only if the adverse selection problem is severe enough, implying that different types' incentives are not too closely aligned.

not enter the market in equilibrium, and for simplicity, suppose that all other agents do enter the market. The rest of the arguments go through even if there are other agents who exit the market. Since i exits the market, its layer is empty, $\tilde{q}_i = \tilde{q}_{i-1}$, and

$$\tilde{T}(\tilde{q}_{i+1}) - \tilde{T}(\tilde{q}_{i-1}) = \tilde{c}_{i+1} (\tilde{q}_{i+1} - \tilde{q}_{i-1}).$$

Now compare that to a hypothetical equilibrium with the same quantities $(\tilde{q}_j)_{j < i}$ and corresponding market tariff $\tilde{T}'$ in which type i enters the market. Hence, $\tilde{q}_i > \tilde{q}_{i-1}$ and

$$\tilde{T}'(\tilde{q}_{i+1}) - \tilde{T}'(\tilde{q}_{i-1}) = \tilde{c}_{i+1} (\tilde{q}_{i+1} - \tilde{q}_i) + \tilde{c}_i (\tilde{q}_i - \tilde{q}_{i-1}) < \tilde{c}_{i+1} (\tilde{q}_{i+1} - \tilde{q}_{i-1}),$$

implying that $i + 1$ will have higher utility with these quantities and market tariff. The same logic that utilities are increasing when lower types are present applies to all agents greater than i . Because their utilities in the market increase, agents with type greater than i are more likely to enter the market when i enters the market. Therefore, an entry of a lower type agent generates one-way strategic complementarity to higher type agents.

However, the effect can go the other way for agent j with type lower than i , i.e. $j < i$. If the upper-tail conditional expected cost of agent j excluding i is less than $\tilde{c}_i$ —that is,

$$\frac{\tilde{T}(\tilde{q}_j) - \tilde{T}(\tilde{q}_{j-1})}{\tilde{q}_j - \tilde{q}_{j-1}} = \frac{\sum_{k \geq j, k \neq i} m_k \tilde{c}_k}{\sum_{k \geq j, k \neq i} m_k} < \tilde{c}_i,$$

then the entry of i will increase the upper-tail conditional expected cost for agents with type j and lower, as the new upper-tail conditional expected cost is

$$\frac{\sum_{k \geq j, k \neq i} m_k}{\sum_{k \geq j} m_k} \frac{\sum_{k \geq j, k \neq i} m_k \tilde{c}_k}{\sum_{k \geq j, k \neq i} m_k} + \frac{m_i}{\sum_{k \geq j} m_k} \tilde{c}_i,$$

which is a convex combination of $\frac{\tilde{T}(\tilde{q}_j) - \tilde{T}(\tilde{q}_{j-1})}{\tilde{q}_j - \tilde{q}_{j-1}}$ and $\tilde{c}_i$, so

$$\frac{\tilde{T}(\tilde{q}_j) - \tilde{T}(\tilde{q}_{j-1})}{\tilde{q}_j - \tilde{q}_{j-1}} = \frac{\sum_{k \geq j, k \neq i} m_k \tilde{c}_k}{\sum_{k \geq j, k \neq i} m_k} < \frac{\sum_{k \geq j, k \neq i} m_k}{\sum_{k \geq j} m_k} \frac{\sum_{k \geq j, k \neq i} m_k \tilde{c}_k}{\sum_{k \geq j, k \neq i} m_k} + \frac{m_i}{\sum_{k \geq j} m_k} \tilde{c}_i.$$

Because of this cost increase, agent j 's maximum utility in the market decreases, and j may exit the market. Hence, entry of higher types can generate one-way strategic substitutability to a lower type. Therefore, the overall effect of the equilibrium remains ambiguous. Thus, the outside options generate key interactions between entry/exit decisions of different agents that can destabilize the otherwise very robust and stable entry-proof equilibrium of [Attar et al. \(2021\)](#). [Levy and Veiga \(2021\)](#) show a similar result in a competitive exclusive contracting environment when the cost c_n is unbounded (or tends to infinity). By contrast, with non-exclusive contracting, our non-existence of equilibrium does not require infinite or unbounded costs.

However, depending on distributional assumptions, outside options also generate the cascade of exits from a small change in fundamentals as in subsection 5.1. Specifically, condition (13) generates Proposition 2, which is important because an exit of the lowest (best) type agent always has an unambiguous, negative, and strategic effect to higher type agents—exits of the lower type agents generate additional exits. Therefore, an additional assumption that provides structure to the distribution of the outside option values is needed to guarantee that the equilibrium exists. Corollary 1 shows that condition (13) restores existence and uniqueness. The oscillation described above therefore requires that the condition fail.

C.3 Generalized Outside Options in No-Trade Equilibrium of [Attar et al. \(2014\)](#)

This subsection examines the role of generalized outside options in the no-trade (pooling) equilibrium of [Attar et al. \(2014\)](#)—that is, both types trade the no-trade contract $(0, 0)$. Proposition C.1 shows that we can revive trade when we generalize outside

options, but Proposition C.2 shows that such a revived equilibrium may go back to no-trade equilibrium after a small decrease in v_L .

Proposition C.1 *Suppose informed types have non-trivial outside options equal to $\gamma_i > u_i(0, 0)$, $i = L, H$. Suppose that the preferences of both types are as in the case of pooling equilibrium in Theorem 1 of Attar et al. (2014). Then, the equilibrium is separating equilibrium if either*

1. $\gamma_L < u_L(\tilde{Q}_L, \tilde{Q}_L v_L)$ and $\gamma_H \geq u_H(\tilde{Q}_L, \tilde{Q}_L v_L)$, where $\tilde{Q}_L < 0$ such that $\tau_L(\tilde{Q}_L, \tilde{Q}_L v_L) = v_L$; or
2. $\gamma_H < u_H(\tilde{Q}_H, \tilde{Q}_H v_H)$ and $\gamma_L \geq u_L(\tilde{Q}_H, \tilde{Q}_H v_H)$, where $\tilde{Q}_H > 0$ such that $\tau_H(\tilde{Q}_H, \tilde{Q}_H v_H) = v_H$.

In all other cases, the equilibrium remains to be pooling equilibrium in which both types choose the outside option.

Proof of Proposition C.1.

Step 1.

Suppose that both types L and H enter the market. Then, the only available contract is the no-trade contract with $(0, 0)$. Hence, both types prefer the outside option over staying in the market. Therefore, there is no equilibrium in which both types enter the market.

Step 2.

Case 1. Suppose that $\gamma_L < u_L(\tilde{Q}_L, \tilde{Q}_L v_L)$, where $\tilde{Q}_L < 0$ is such that $\tau_L(\tilde{Q}_L, \tilde{Q}_L v_L) = v_L$. Suppose that H takes the outside option and L remains in the market. Then, $v_L \leq \tau_L(0, 0)$ implies that L type would buy at the price v_L , i.e., $\tilde{Q}_L < 0$.

Case 1.1. If $\gamma_H \geq u_H(\tilde{Q}_L, \tilde{Q}_L v_L)$, then equilibrium is separating with $\tilde{Q}_L < 0$, $\tilde{Q}_H = 0$, because H type would prefer to take the outside option over the market trade.

Case 1.2. If $\gamma_H < u_H(\tilde{Q}_L, \tilde{Q}_L v_L)$, then H type also prefers to trade the contract $(\tilde{Q}_L, \tilde{Q}_L v_L)$ in the market and would like to enter. However, if both types enter the market, then Theorem 1 in Attar et al. (2014) implies that the only possible equilibrium contract is $(0,0)$, which cannot be an equilibrium as we discussed in Step 1. Therefore, there is no equilibrium with H type exits and L type enters, if $\gamma_L < u_L(\tilde{Q}_L, \tilde{Q}_L v_L)$ and $\gamma_H < u_H(\tilde{Q}_L, \tilde{Q}_L v_L)$.

Case 2. Suppose that $\gamma_H < u_H(\tilde{Q}_H, \tilde{Q}_H v_H)$, where $\tilde{Q}_H > 0$ is such that $\tau_H(\tilde{Q}_H, \tilde{Q}_H v_H) = v_H$. Suppose that L type exits and H remains in the market. Then, $\tau_H(0,0) \leq v_H$ implies that H type would sell at the price v_H , i.e., $\tilde{Q}_H > 0$.

Case 2.1. If $\gamma_L \geq u_L(\tilde{Q}_H, \tilde{Q}_H v_H)$, then equilibrium is separating with $Q_L = 0, \tilde{Q}_H > 0$, because L type would prefer to take the outside option over the market trade.

Case 2.2. If $\gamma_L < u_L(\tilde{Q}_H, \tilde{Q}_H v_H)$, then L type also prefers to trade the contract $(\tilde{Q}_H, \tilde{Q}_H v_H)$ in the market and would like to enter. However, if both types enter the market, then Theorem 1 in Attar et al. (2014) implies that the only possible equilibrium contract is $(0,0)$, which cannot be an equilibrium as we discussed in Case 1. Therefore, there is no equilibrium with H type enters and L type exits, if $\gamma_H < u_H(\tilde{Q}_H, \tilde{Q}_H v_H)$ and $\gamma_L < u_L(\tilde{Q}_H, \tilde{Q}_H v_H)$.

Case 3. If $\gamma_H \geq u_H(\tilde{Q}_H, \tilde{Q}_H v_H)$ and $\gamma_L \geq u_L(\tilde{Q}_L, \tilde{Q}_L v_L)$, then both L and H type exit the market. ■

Proposition C.1 implies that we can revive trades in the market even in the case in which there were no trades in Attar et al. (2014). Outside options can be effective in screening out one type from mimicking the other type, as the utility from the outside options can exceed the payoff from mimicking the market trade of the other type. However, this result depends on the values of the outside options.

One last thing to note is that there could be self-fulfilling no trade equilibrium. Even if $\gamma_L < u_L(\tilde{Q}_L, \tilde{Q}_L v_L)$ and $\gamma_H < u_H(\tilde{Q}_H, \tilde{Q}_H v_H)$, if the uninformed investors believe that both types are exiting, they are indifferent between offering any contract,

including the no-trade contract $(0, 0)$. If the no-trade contract is the only contract being offered, both types would prefer to exit and take their outside option. However, we can rule out this self-fulfilling no-trade equilibrium by simple equilibrium refinements such as the trembling-hand equilibrium or sequential equilibrium. For the simplicity of exposition, we do not go into this equilibrium refinement and assume that there is no self-fulfilling no-trade equilibrium.

The next proposition shows that such an equilibrium with positive (negative) trades can break down after a small changes in the values (costs) for uninformed buyers, when the utility levels of the outside options are close to the edge of mimicking the other type. Under such a case, the equilibrium goes back to the no-trade equilibrium.

Proposition C.2 *Suppose that the conditions in Proposition C.1 hold. A small decrease in v_L can make an equilibrium with positive trades to breakdown to a no-trade equilibrium.*

Proof.

Case 1. Suppose that $\gamma_L < u_L(\tilde{Q}_L, \tilde{Q}_L v_L)$, where $\tilde{Q}_L < 0$ is such that $\tau_L(\tilde{Q}_L, \tilde{Q}_L v_L) = v_L$. Now consider the case of v_L decreasing to $\hat{v}_L$.

Case 1.1. Suppose that the equilibrium is a separating equilibrium because L type enters, while H type exits, i.e., $\gamma_L < u_L(\tilde{Q}_L, \tilde{Q}_L v_L)$ and $\gamma_H \geq u_H(\tilde{Q}_L, \tilde{Q}_L v_L)$. If v_L decreases to $\hat{v}_L$, then L type still prefers to trade in the market with $\hat{\tilde{Q}}_L < \tilde{Q}_L$ such that $\tau_L(\hat{\tilde{Q}}_L, \hat{\tilde{Q}}_L \hat{v}_L) = \hat{v}_L$. However, the inequality for the H type can be reversed as $u_H(\hat{\tilde{Q}}_L, \hat{\tilde{Q}}_L \hat{v}_L) > \gamma_H \geq u_H(\tilde{Q}_L, \tilde{Q}_L v_L)$. Thus, H type now has an incentive to mimic the L type, implying that the uninformed buyers will only suggest no trade contracts, leading to both types choosing the outside option over the market trade.

Case 1.2. Suppose that the equilibrium is a pooling equilibrium with both types choosing the outside option because $\gamma_L < u_L(\tilde{Q}_L, \tilde{Q}_L v_L)$ and $\gamma_H < u_H(\tilde{Q}_L, \tilde{Q}_L v_L)$, i.e., the H type would like to mimic the L type if there is a contract designed for

the L type (Case 1.2. in the proof of Proposition C.1). Then, the unit price of the contract $\hat{v}_L$ is lower than before, so both inequalities remain the same as

$$\begin{aligned}\gamma_L &< u_L(\hat{\tilde{Q}}_L, \hat{\tilde{Q}}_L \hat{v}_L) \\ \gamma_H &< u_H(\hat{\tilde{Q}}_L, \hat{\tilde{Q}}_L \hat{v}_L),\end{aligned}$$

therefore, the equilibrium is still a pooling equilibrium with both types choosing the outside option.

Case 2. Suppose that $\gamma_H < u_H(\tilde{Q}_H, \tilde{Q}_H v_H)$, where $\tilde{Q}_H > 0$ is such that $\tau_H(\tilde{Q}_H, \tilde{Q}_H v_H) = v_H$. Now consider the case of v_L decreasing to $\hat{v}_L$.

In this case, the equilibrium contract $(\tilde{Q}_H, \tilde{Q}_H v_H)$ and every payoff relevant to the equilibrium, $u_H(\tilde{Q}_H, \tilde{Q}_H v_H)$, the L type's mimicking payoff $u_L(\tilde{Q}_H, \tilde{Q}_H v_H)$, and both outside options, are independent of v_L . A small decrease in v_L therefore leaves the separating equilibrium unchanged.

Case 3. In a knife-edge case of Case 3 in the proof of Proposition C.1, $\gamma_L = u_L(\tilde{Q}_L, \tilde{Q}_L v_L)$, a decrease in v_L can make the equilibrium a separating equilibrium if $\gamma_H \geq u_H(\hat{\tilde{Q}}_L, \hat{\tilde{Q}}_L \hat{v}_L)$, i.e. L type purchases the good in the market, and H type chooses the outside option. ■

D Numerical Example

This section provides a numerical example to highlight the main analytical results. In particular, consider the canonical insurance market with binary loss model of [Rothschild and Stiglitz \(1976\)](#) and [Hendren \(2014\)](#) following the idea of [Dubey and Geanakoplos \(2019\)](#).²⁷ We first compare the results between the baseline model of [Attar et al. \(2021\)](#) with and without outside options. Then, we consider the comparative statics on the partition of types.

D.1 Model Setup and Equilibrium

Agents have initial endowment $(e, 0)$ for (e_g, e_b) , where e_g and e_b represent good state and bad state endowment, respectively. They will consume $(x_g, x_b) = (e, 0)$ under autarky, where x_g and x_b represent good state and bad state consumption, respectively. Suppose that agents have constant relative risk aversion (CRRA) and agent i 's utility function is

$$v_i(x_g, x_b) = p_i \log(1 + x_b) + (1 - p_i) \log(1 + x_g),$$

where p_i is the probability of agent i faces a loss and receives bad state endowment. The marginal rate of substitution for agent i is

$$\tau_i(x_g, x_b) \equiv \frac{\frac{\partial v_i(x_g, x_b)}{\partial x_b}}{\frac{\partial v_i(x_g, x_b)}{\partial x_g}} = \frac{p_i}{1 - p_i} \frac{1 + x_g}{1 + x_b}.$$

Following the assumption of competitive suppliers in the main model, the service cost for suppliers is $c_i = p_i$, so they need p_i amount of x_g to insure $1 - p_i$ amount of x_b to agent i . In the notation of the main model, q denotes gross insurance indemnity and

²⁷Note that the baseline model of [Attar et al. \(2021\)](#) and our extension with outside options in this paper are much more general and can be applied to many other contexts.

t denotes the premium. Hence, state-contingent consumption satisfies

$$\begin{aligned}x_g &= e - t \\x_b &= q - t.\end{aligned}$$

Since the supplier's expected cost of gross indemnity q is $p_i q$, zero profit requires $t = p_i q$, so the unit cost in (q, t) space is $c_i = p_i$. Thus, eliminating q yields

$$e - x_g = t = \frac{p_i}{1 - p_i} x_b,$$

which explains why the corresponding actuarially fair frontier in (x_g, x_b) space has a slope of $p_i/(1 - p_i)$.

Under non-exclusive contracts, suppliers require the upper-tail conditional expected cost,

$$\bar{c}_i = \sum_{j \geq i} \frac{m_j p_j}{\sum_{j \geq i} m_j},$$

where m_i is the relative mass of agent i .

Denote the additional utility level of taking the outside option on top of utility under no trade, $v_i(e, 0)$, as γ for each i . Agents will compare the level of utility they can get from their optimal consumption bundle in the market to the level of utility they can get from the outside option and decide whether to enter or exit the market. Since $v_i(e, 0) = (1 - p_i) \log(1 + e)$ is decreasing in i , the specification $\gamma_i \equiv v_i(e, 0) + \gamma$ satisfies condition (13) for all market bundles.²⁸ The algorithm below implements the sequential exit process of Definition 2. By Corollary 1, its terminal set is the unique equilibrium active set.

²⁸For any bundle with $x_b > 0$ and $x_g \leq e$, $(v_j(x_g, x_b) - \gamma_j) - (v_i(x_g, x_b) - \gamma_i) = (p_j - p_i) \log[(1 + x_b)(1 + e)/(1 + x_g)] > 0$ for $j > i$, so $v_i \geq \gamma_i$ implies $v_j > \gamma_j$. All bundles arising in the analysis involve nonnegative transfers, hence $x_g \leq e$.

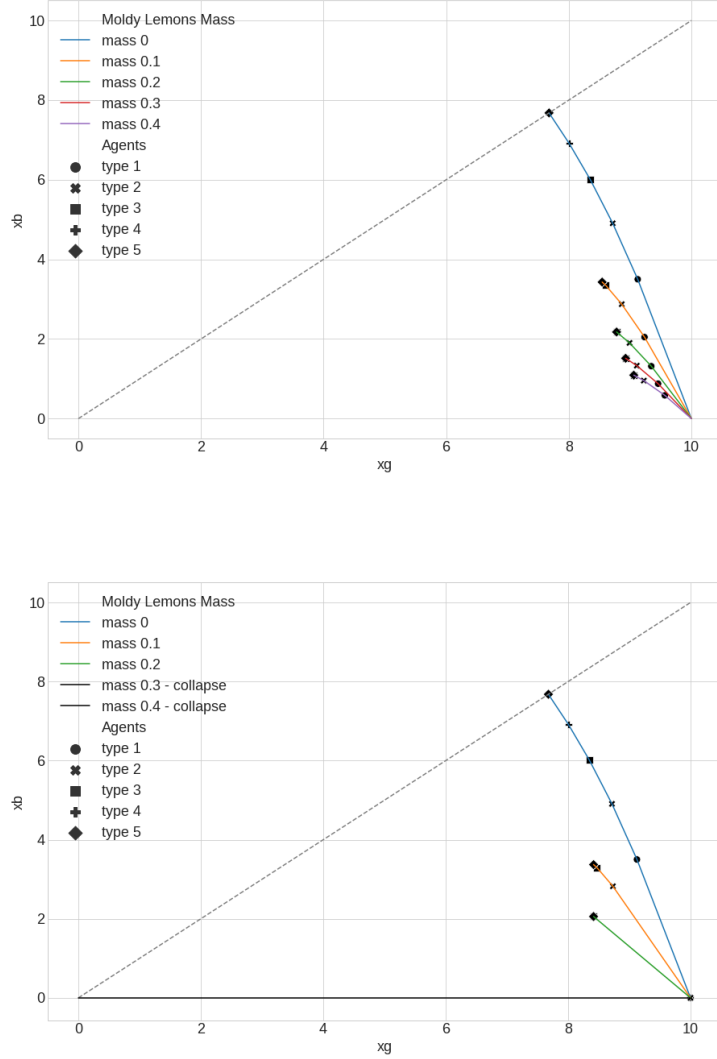


Figure D.1: Consumption bundles of models without and with outside options
Note: Each curve represents consumption possibility frontier and consumption bundles of each agent represented by different shape of dot for a different mass of moldy lemons (the new worst type agent).

D.2 Effect of Outside Options and Cost Increases

We verify the main result of this paper with the numerical model. The cost increase we consider arises from entry of a worst type of agent, which we call a “moldy lemon”.

This cost increase is a natural case to consider and highlights the amplification result. Denote the cost of supplying moldy lemons as c_{n+1} and the mass of moldy lemons as m_{n+1} . Figure D.1 shows equilibrium of the baseline model in the top panel and equilibrium of the model with outside options in the bottom panel. For each panel, the horizontal axis represents consumption in the good state, x_g , and the vertical axis represents consumption in the bad state, x_b . Each colored curve represents the consumption possibility frontier for a given mass of moldy lemons in the market. Different shapes of dots represent each agent's optimal consumption bundle given mass of moldy lemons and other agents' consumption in the equilibrium.

First, the results show that x_b decreases monotonically with the increase in the mass of moldy lemons. As more moldy lemons enter the market, the upper-tail conditional expected cost $\tilde{c}_i$ increases, depressing the consumption possibility frontier for every agent in the market. With the higher cost, agents purchase less (insurance) contracts.²⁹

Second, the results show a market shutdown in the model with outside options, which does not exist in the model without outside options. Even though total quantities decrease in both models, the model without outside options shows a smooth contraction of quantities traded as the mass of moldy lemons rises. Thus, even when the moldy lemons mass is 0.4, all agents still enter the market and trade in positive amounts. In contrast, the model with outside option shows the exit of agents and total market shutdown at the moldy lemons mass of 0.3. Thus, the numerical exercise illustrates how the existence of outside options can generate complete market shutdowns, which Theorem 2 characterizes and Corollary 2 shows can be triggered by a small cost increase once the initially marginal type is sufficiently close to its participation threshold.³⁰

²⁹This decrease in quantities may not always be the case in the general model, because the income effects might dominate the substitution effects. The functional form of this exercise precludes the case of insurance being a Giffen good.

³⁰The definition of complete market shutdown here is the exit of all agents except for moldy lemons.

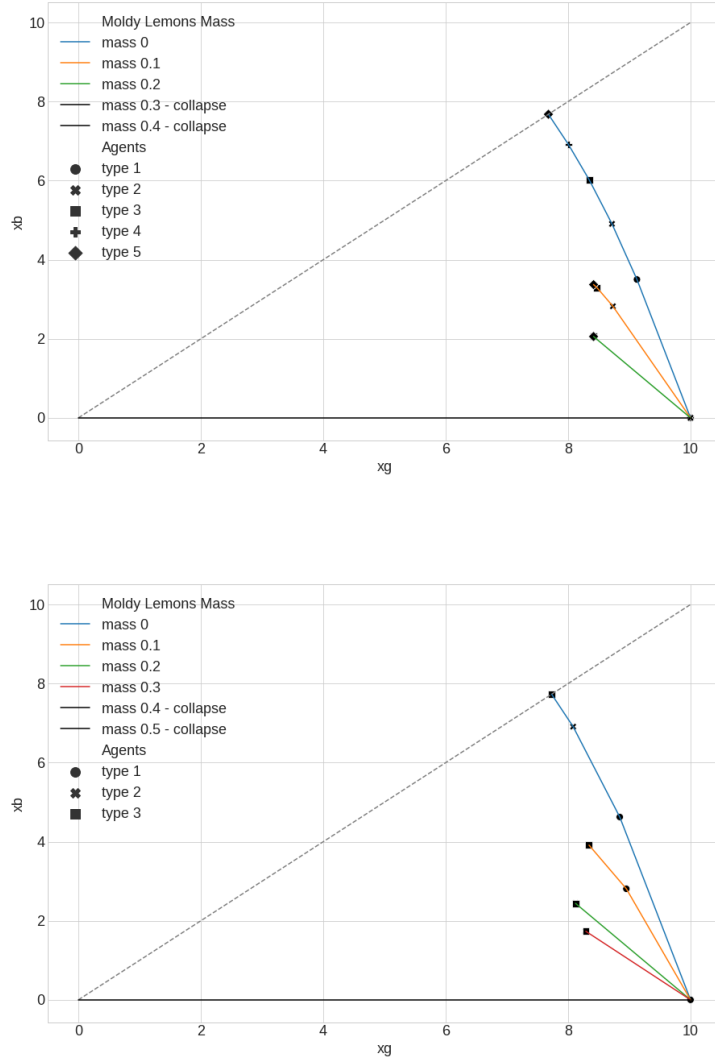


Figure D.2: Consumption bundles of models with different partition of types
Note: Each curve represents consumption possibility frontier and consumption bundles of each agent represented by different shape of dot for a different mass of moldy lemons.

D.3 Coarse Partition of Types and Cost Increases

Using the model with outside options, we analyze comparative statics of Proposition 5 on the division of types depicted in Figure D.2. In particular, we group adjacent types of agents into a single type whose service cost is the weighted average of individual

type’s cost, and a mass equal to the sum of each type’s mass. In particular, we combine type 1 and 2 together to create a new type 1 agent with probability and mass as

$$\hat{p}_1 = \frac{m_1 p_1 + m_2 p_2}{m_1 + m_2}$$

$$\hat{m}_1 = m_1 + m_2.$$

Similarly, we also combine type 3 and 4 in the baseline case into a new type 2 agent as

$$\hat{p}_2 = \frac{m_3 p_3 + m_4 p_4}{m_3 + m_4}$$

$$\hat{m}_2 = m_3 + m_4,$$

while simply renaming the previous agent 5 as agent 3. Therefore, the market-wide uncertainty and service cost remain the same as before.

The top panel of Figure D.2 is the baseline case as shown in the bottom panel of Figure D.1, while the bottom panel of Figure D.2 is the case with a more coarse partition. The numerical results show that the economy with a coarser partition of types is less vulnerable due to cost increases arising from the entry of moldy lemons—there are fewer exits across varying masses of moldy lemons. Also, full market shutdown does not happen even when the moldy lemons mass is 0.3, which generated a full market collapse under the baseline case.

D.4 Details of Numerical Simulations

For computational tractability, we would like to represent agents’ optimization problem as isomorphic optimization problem across all agents by adjusting the endowments e^i for each agent i . This can be done by exploiting the single-crossing property and any agent $i > j$ would consume as much as agent j does in equilibrium. For a

given e^i , the optimal consumption bundle is

$$\begin{aligned} x_g^* &= (1 - p_i)e^i - \frac{p_i - \bar{c}_i}{1 - \bar{c}_i} \\ x_b^* &= \frac{p_i - \bar{c}_i}{\bar{c}_i} + \frac{(1 - \bar{c}_i)p_i}{\bar{c}_i}e^i, \end{aligned}$$

for an interior solution. If it is a corner solution, then $(x_g^*, x_b^*) = \left(0, \frac{1 - \bar{c}_i}{\bar{c}_i}e^i\right)$ or $(x_g^*, x_b^*) = (e^i, 0)$.

From the results in the general model, we know that agent 1 first decides on the optimal quantity q_1^* for the given price $\bar{c}_1$ and then agent 2 decides on q_2^* for the price $\bar{c}_2$ on top of purchasing q_1^* , and so forth. Agent 1's optimal consumption bundle is

$$\begin{aligned} x_g^1 &= (1 - p_1)e - \frac{p_1 - \bar{c}_1}{1 - \bar{c}_1} \\ x_b^1 &= \frac{p_1 - \bar{c}_1}{\bar{c}_1} + \frac{(1 - \bar{c}_1)p_1}{\bar{c}_1}e, \end{aligned}$$

assuming that they have interior solutions without loss of generality. For agent 2, the same optimality condition should hold, but the budget constraint is different from the previous representation. This is because agent 2 can purchase the bundle in a cheaper price $\frac{\bar{c}_1}{1 - \bar{c}_1}$ instead of $\frac{\bar{c}_2}{1 - \bar{c}_2}$ up to x_b^1 . Therefore, the updated budget constraint for agent 2 becomes

$$\begin{aligned} x_g^2 &= e - \frac{\bar{c}_1}{1 - \bar{c}_1}x_b^1 - \frac{\bar{c}_2}{1 - \bar{c}_2}(x_b^2 - x_b^1) \\ &= e + \left(\frac{\bar{c}_2}{1 - \bar{c}_2} - \frac{\bar{c}_1}{1 - \bar{c}_1}\right)x_b^1 - \frac{\bar{c}_2}{1 - \bar{c}_2}x_b^2. \end{aligned}$$

Therefore, we simply change the endowment from e to

$$e^2 = e + \left(\frac{\bar{c}_2}{1 - \bar{c}_2} - \frac{\bar{c}_1}{1 - \bar{c}_1}\right)x_b^1$$

for the agent 2's budget constraint. Thus, the optimal consumption bundle for agent

2 is

$$\begin{aligned} x_g^2 &= (1 - p_2)e^2 - \frac{p_2 - \bar{c}_2}{1 - \bar{c}_2} \\ x_b^2 &= \frac{p_2 - \bar{c}_2}{\bar{c}_2} + \frac{(1 - \bar{c}_2)p_2}{\bar{c}_2} e^2. \end{aligned}$$

Agent 3's problem is isomorphic to agent 2's problem except that agent 3's endowment is

$$e^3 = e + \left(\frac{\bar{c}_2}{1 - \bar{c}_2} - \frac{\bar{c}_1}{1 - \bar{c}_1} \right) x_b^1 + \left(\frac{\bar{c}_3}{1 - \bar{c}_3} - \frac{\bar{c}_2}{1 - \bar{c}_2} \right) x_b^2,$$

and agent 3's optimal consumption bundle becomes

$$\begin{aligned} x_g^3 &= (1 - p_3)e^3 - \frac{p_3 - \bar{c}_3}{1 - \bar{c}_3} \\ x_b^3 &= \frac{p_3 - \bar{c}_3}{\bar{c}_3} + \frac{(1 - \bar{c}_3)p_3}{\bar{c}_3} e^3. \end{aligned}$$

For a general agent i , agent i 's updated endowment is

$$e^i = e + \sum_{j < i} \left(\frac{\bar{c}_{j+1}}{1 - \bar{c}_{j+1}} - \frac{\bar{c}_j}{1 - \bar{c}_j} \right) x_b^j,$$

and agent i 's optimal consumption bundle becomes

$$\begin{aligned} x_g^i &= (1 - p_i)e^i - \frac{p_i - \bar{c}_i}{1 - \bar{c}_i} \\ x_b^i &= \frac{p_i - \bar{c}_i}{\bar{c}_i} + \frac{(1 - \bar{c}_i)p_i}{\bar{c}_i} e^i. \end{aligned}$$

Given these setup, the parameters of the model are as in the following Table [D.1](#).

For the numerical procedure, we apply the exit test of the sequential exit process (Definition [2](#)) starting from $i = 1$ and updating the endowment of $e^{i+1} = e$ whenever the condition is satisfied. The algorithm repeats this until it finds the agent that

Parameter	Description	Value
e	good state endowment	10
$(p_1, p_2, \dots, p_5)$	probability of bad state	$(0.1, 0.15, \dots, 0.3)$
$(m_1, m_2, \dots, m_5)$	mass of each type	$(0.2, 0.2, \dots, 0.2)$
γ	outside option utility level	0.0758

Table D.1: Parameter values for numerical simulations

does not violate the test. Then, we proceed with the rest of the agents' consumption quantities using the iterative representation.

The algorithm that solves this numerical model is the following: For each i ,

1. Calculate the endowment of agent i , e^i , using the previous agents' quantities.
2. Derive the optimal quantity for agent i , (x_g^i, x_b^i) under e^i and $\bar{c}_i$.
3. Compute the utility level $v_i(x_g^i, x_b^i)$ and compare that to $v_i(e, 0) + \gamma$.
4. If it is above $v_i(e, 0) + \gamma$, then the (x_g^i, x_b^i) quantity is the optimal quantity. Otherwise, set $(x_g^i, x_b^i) = (e, 0)$.
5. Move to the next agent $i + 1$ and repeat from the first step.